

**Planning and Engineering of Water and Sewage
Treatment Plants at Washnah Contracting
Focused on WTP & STP Operations, Hydraulic
Calculations, and Progress Tracking.**

Summer internship Report

By

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2024 – 2025

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Washnah Contracting for the opportunity to intern and learn under their guidance. This experience allowed me to develop valuable insights into Water Treatment Plant (WTP) and Sewage Treatment Plant (STP) operations, particularly in areas such as treatment processes, hydraulic profiling, pipe sizing, and planning engineering.

I am deeply thankful to my supervisor and the entire team at Washnah Contracting for their mentorship and support throughout my internship. Their expertise and encouragement enabled me to enhance my technical skills, particularly in project scheduling and management using Oracle Primavera P6, as well as in drafting process flow diagrams in AutoCAD.

I would also like to thank my colleagues for their collaboration and willingness to share their knowledge, which greatly enriched my understanding of practical engineering applications. This internship has been an invaluable learning experience, and I am grateful to have had the chance to work in such a supportive and professional environment.

I am deeply indebted to my mentor at VJTI Mumbai, Dr Arvind Deshpande for his constant encouragement and insightful feedback throughout this journey.

Mohammed Wafeeq Mohd. Salim Kazi

ABSTRACT

The purpose of this internship at Washnah Contracting was to build a comprehensive foundation in both the technical and project management aspects of Water Treatment Plant (WTP) and Sewage Treatment Plant (STP) operations.

By immersing in the day-to-day processes and challenges of WTP/STP projects, this internship aimed to equip me with the skills required for effective design, planning, and execution of water and wastewater treatment facilities.

Key objectives included understanding the intricate stages and technologies involved in water treatment, mastering calculations for head loss and pipe sizing essential for plant efficiency, and enhancing my ability to draft detailed process flow diagrams using AutoCAD.

Additionally, learning to interpret hydraulic profiles enabled me to grasp how different plant sections influence flow and pressure distribution, ensuring operational consistency within design parameters.

Furthermore, proficiency in Oracle Primavera P6 was emphasized to develop project schedules, allocate resources, and monitor progress meticulously. Creating sequential activity lists for smooth project execution and generating detailed progress reports allowed me to communicate effectively with stakeholders, ensuring alignment and timely project completion.

This experience has provided a strong technical and managerial skill set critical for future contributions to the water treatment sector.

DECLARATION

I, Mohammed Wafeeq Mohd. Salim Kazi, hereby declare that the report titled " **Planning and Engineering of Water and Sewage Treatment Plants at Washnah Contracting Focused on WTP & STP Operations, Hydraulic Calculations, and Progress Tracking** " is the result of my own work carried out during my internship at Washnah, Jeddah, Saudi Arabia. This report has not been submitted to any other institution or organization for any certificate or any qualification.

All the information and data presented in this report have been obtained and presented in accordance with academic rules and ethical conduct. I have acknowledged all sources of information and data from other works in the references section of this report.

I take full responsibility for the contents of this report and attest that the information contained herein is accurate and complete to the best of my knowledge.

Mohammed Wafeeq Mohd. Salim Kazi

(211020022)

Date:

Place: VJTI, Mumbai

CERTIFICATE

This is to certify that the report titled " **Planning and Engineering of Water and Sewage Treatment Plants at Washnah Contracting Focused on WTP & STP Operations, Hydraulic Calculations, and Progress Tracking** " by **Mohammed Wafeeq Mohd. Salim Kazi** has been found satisfactory as part of the internship at Washnah, Jeddah, Saudi Arabia.

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Place: VJTI, Mumbai

Table Of Contents

1	About The Company	8
2	Types of plants	9
2.1	Water Treatment Plants	9
2.2	Sewage Treatment Plants	10
3	Plant Layout.....	11
3.1	Process Flow diagram	11
3.2	Hydraulic profile drawing.....	13
3.3	Layout Plan of Plant	15
4	Equipment's used in WTP/STP in brief	17
4.1	Inlet Chamber.....	17
4.2	Grit And Coarse Screen.....	17
4.3	Lift And Pump Station.....	18
4.4	Air Blowers	18
4.5	Aeration Tanks	19
4.6	Clarifier Tanks	19
4.7	Gravity Filters	20
4.8	Flowmeter and Monitoring Equipment's	20
4.9	Activated Carbon or Odor Control Systems	21
4.10	Chlorination Chambers	21
4.11	Sludge Handling Systems.....	22
4.12	Backwash Pump and Wash Water Tank	22
4.13	Chemical Dosing Systems	23
4.14	UV Chambers	23
4.15	Submersible Pumps	24
4.16	Return Sludge Pumps.....	24
4.17	Anoxic Chamber.....	25
5	Some Design Calculations.....	26

5.1	Head Loss Calculation _____	26
5.2	Wier And its Calculation _____	30
5.3	Pipe Sizing Calculation _____	34
6	Project Management Through Primavera P6.....	36
6.1	Key Features of PRIMAVERA P6 _____	36
6.2	Gantt chart and Activity Lists _____	38
6.3	Resource Allocation _____	50
6.4	Scheduling / Levelling Report _____	57
7	Appendix	73
7.1	Internship confirmation _____	73
7.2	Internship certificate _____	74
7.3	References _____	75

1 About The Company

Washnah Trading and Contracting, founded in 1992, is in Saudi Arabia's water treatment and wastewater management industry. It specializes in providing solutions for Water Treatment Plants (WTP) and Sewage Treatment Plants (STP), catering to the public and private sectors. With a strong focus on sustainability and innovative technologies, the company addresses essential water management challenges within the Kingdom.



➤ **Water and Wastewater Treatment Solutions:**

Washnah excels in designing, engineering, constructing, and maintaining water and sewage treatment facilities. Its projects span residential, commercial, and industrial applications, helping manage the water needs of urban developments, municipalities, and industrial sites.

➤ **Customized Engineering Solutions:**

Washnah tailors its water treatment systems to meet the specific needs of each client, ensuring solutions align with environmental standards, operational goals, and long-term cost-efficiency. Their engineers frequently use advanced software and project management tools, such as Oracle Primavera P6, to streamline project planning, scheduling, and resource allocation.

➤ **Technical Innovation and Sustainability Focus:**

Emphasizing eco-friendly practices, Washnah integrates advanced filtration, biological treatment, and desalination technologies into their projects, supporting Saudi Arabia's sustainability goals and reducing the environmental footprint of its water infrastructure.

2 Types of plants

2.1 Water Treatment Plants

- Water Treatment Plants (WTPs) are facilities dedicated to purifying raw water from natural sources such as rivers, lakes, reservoirs, or groundwater to make it safe for human consumption and other uses. This process removes contaminants like suspended solids, bacteria, viruses, and chemicals, ensuring that the water meets stringent quality standards for potable use.
- At the start of the treatment process, water is collected from the source and passed through screens to remove large debris like leaves, sticks, and trash. Then, chemicals such as alum (aluminium sulphate) are added in a step known as coagulation, which binds small particles together to form larger clusters called flocs.
- These flocs are then more easily removed in the next step, sedimentation, where they settle at the bottom of tanks as sludge.
- After sedimentation, the water undergoes filtration, where it passes through materials like sand, gravel, and activated carbon to remove any remaining particles, bacteria, and organic compounds.
- Disinfection follows, often through chlorination or UV treatment, to kill any remaining pathogens. In some cases, a post-treatment process adjusts the water's pH to make it non-corrosive and ensures the water is safe and stable for distribution.
- Treated water is then stored in reservoirs before being pumped through a distribution network to reach consumers. WTPs are essential for maintaining public health by ensuring that drinking water is free of contaminants and consistently available.

2.2 Sewage Treatment Plants

- Sewage Treatment Plants (STPs) are designed to treat wastewater from homes, commercial buildings, and sometimes industrial facilities, processing sewage that contains organic and inorganic pollutants, pathogens, and other contaminants. The primary objective of an STP is to remove harmful contaminants, ensuring that treated water can be safely discharged into the environment or reused for applications like irrigation, landscaping, or industrial processes.
- The treatment process begins with preliminary treatment, where large debris such as plastic, rags, and other non-biodegradable materials are screened out, and grit (such as sand and small stones) is removed to prevent wear on equipment.
- Following this, in the primary treatment stage, the wastewater enters settling tanks where heavier solids settle at the bottom, forming sludge, while oils and grease float to the surface and are skimmed off. The sludge collected is either treated or disposed of separately. Secondary treatment involves biological processes where aerobic bacteria break down the organic matter in wastewater, usually in activated sludge systems or biological reactors. After biological treatment, the wastewater flows into secondary sedimentation tanks, where additional solids are removed.
- For facilities requiring advanced treatment, a tertiary treatment step follows, which includes finer filtration, disinfection, and sometimes nutrient removal to prevent excess phosphorus or nitrogen from causing environmental harm like algae blooms.
- The resulting sludge from primary and secondary treatment stages is digested, often anaerobically, to stabilize and reduce its volume, and then dewatered for disposal or, if safe, used as fertilizer. It's important to note that the processes may change depending on the quality of water weather its STP or WTP.

3 Plant Layout

3.1 Process Flow diagram

- A Process Flow Diagram (PFD) provides a clear representation of the sequence of operations and interactions within a treatment plant. It illustrates each step of the treatment process, including components like pumps, tanks, valves, and reactors, as well as the flow of water or wastewater between these components.
- The PFD uses standardized symbols and connectors to depict the different treatment stages, giving engineers a quick overview of the treatment plant's entire workflow.
- This diagram is essential in both the design and operational phases, allowing operators to understand the process flow, identify critical points for monitoring, and troubleshoot potential issues.
- For instance, in a Water Treatment Plant (WTP), the PFD might show stages like coagulation, sedimentation, filtration, and disinfection, while in a Sewage Treatment Plant (STP), it would include screening, grit removal, primary and secondary treatment, and sludge management.
- In addition to guiding the operation of a plant, PFDs are crucial for planning plant expansions or upgrades, as they help engineers assess the impact of changes on the process flow.
- Accurate PFDs also support regulatory compliance, as they are often required by regulatory agencies to ensure that the plant meets environmental standards.
- An example of a process flow diagram is given on the next page.

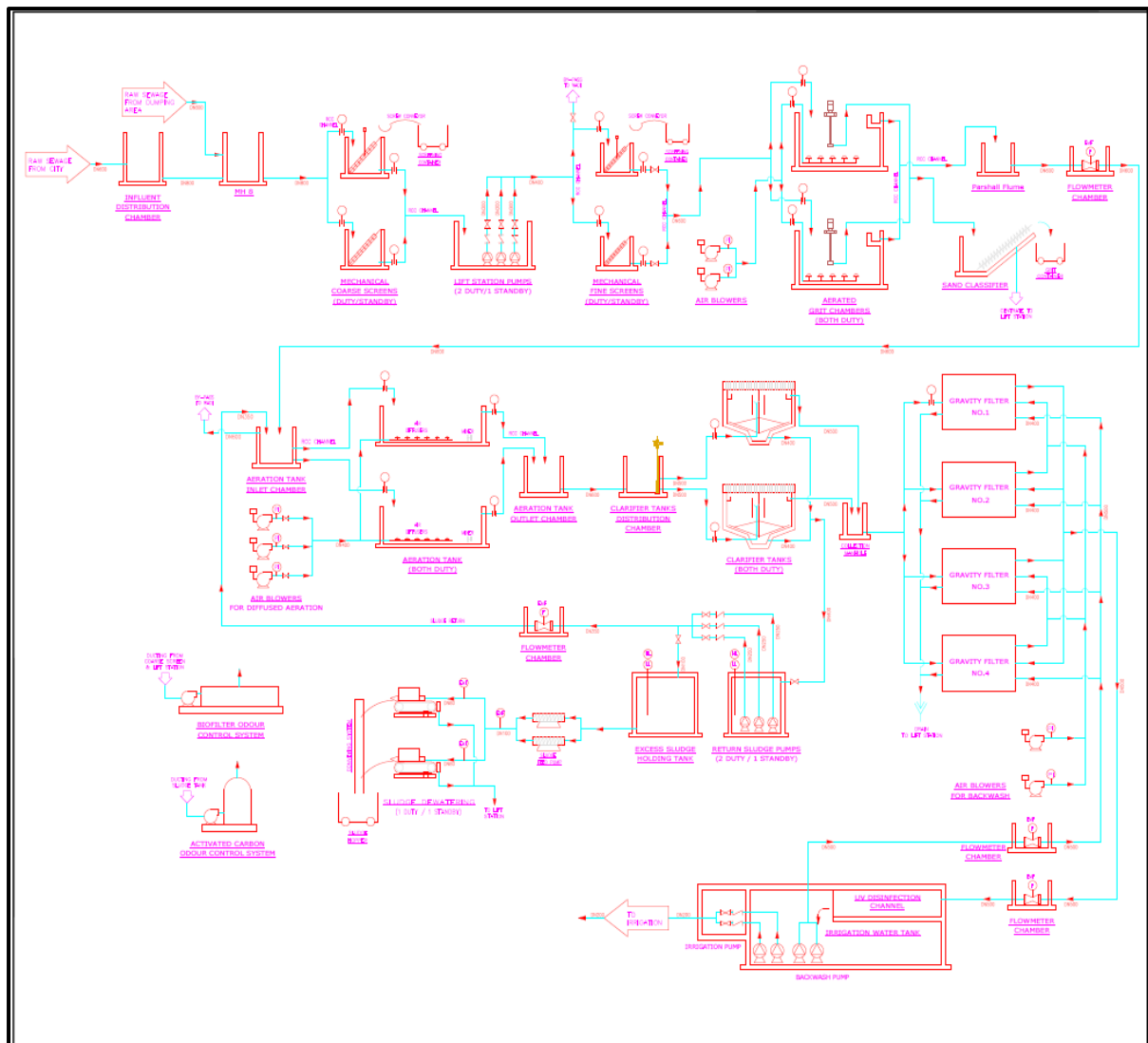


FIG: This image shows a Process flow diagram for a WTP / STP plant.

- Note that the image provided may not be that clear since the original drawing made was on A1 sheet and this report is limited to A4, A3 sheets.

3.2 Hydraulic profile drawing

- The Hydraulic Profile Diagram displays the variation in water levels and pressure heads as water or wastewater moves through the plant. It essentially represents the “height” or elevation of the water at different stages of treatment, allowing engineers to ensure that gravity flow or pumped flow will work as required.
- The diagram is often aligned with the physical layout of the plant to show how water flows through each process and to avoid any instances of backflow or overflow, which could damage the equipment or lead to operational inefficiencies.
- For example, in a WTP, the hydraulic profile would illustrate the changes in water level as it passes from sedimentation to filtration and finally to the storage tank.
- In STPs, the hydraulic profile helps to visualize levels through primary and secondary treatment stages, showing how each treatment component operates within the system’s hydraulic requirements.
- This information is key for ensuring that equipment such as pumps are properly sized to handle head requirements and for identifying any necessary adjustments to avoid flooding or flow disruptions.
- The hydraulic profile is especially critical in facilities using gravity flow, where elevation changes must be carefully managed.
- An example of a process flow diagram is given on the next page.

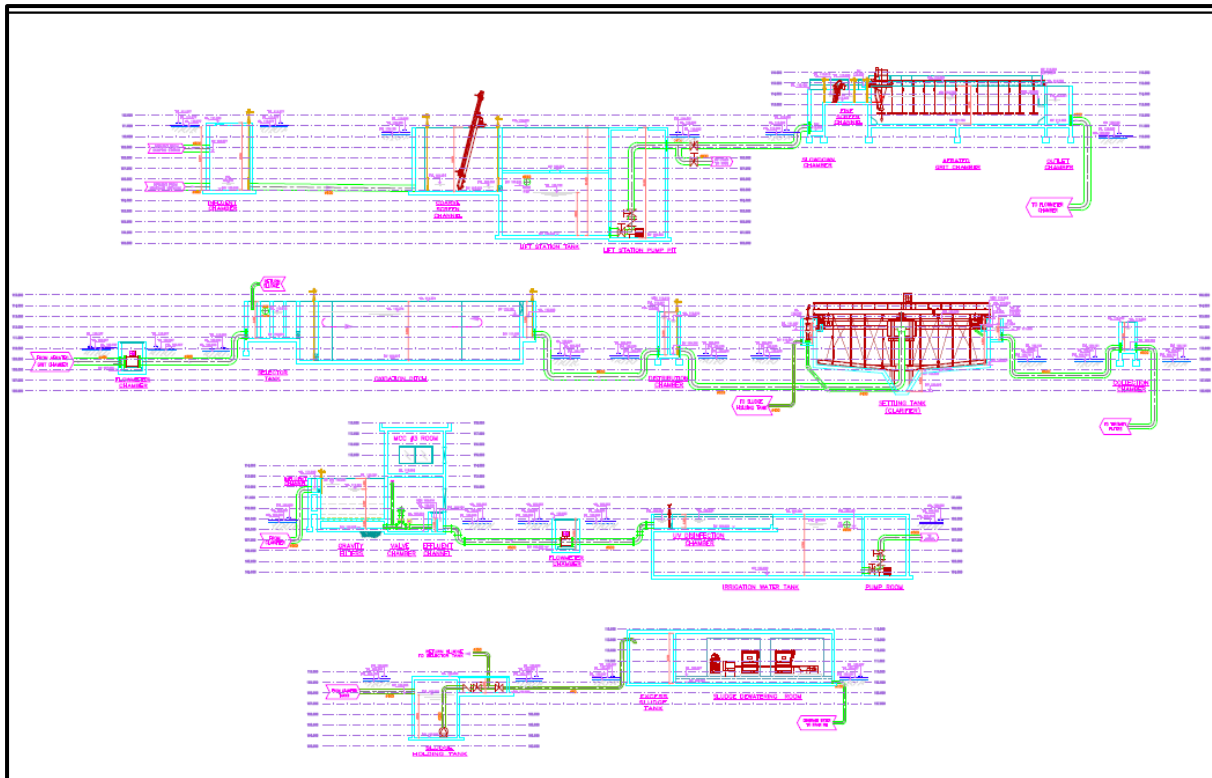


FIG: This image shows hydraulic profile for a WTP / STP plant.

- Note that the image provided may not be that clear since the original drawing made was on A1 sheet and this report is limited to A4, A3 sheets.

3.3 Layout Plan of Plant

- A Layout Map of the plant provides a top-down view of the entire facility, showing the physical placement and dimensions of each treatment component and building within the plant's footprint.
- This map is typically scaled to provide an accurate spatial arrangement, showing the location of each structure, such as intake channels, treatment tanks, sludge handling areas, control rooms, and storage areas, along with access routes, pipelines, and electrical connections.
- The layout map allows engineers and operators to visualize how different components are interconnected spatially, ensuring a logical and efficient arrangement for operations, maintenance, and safety.
- The layout map is also critical for project planning and construction, as it provides contractors with essential information on where to install equipment and where to leave space for future expansions.
- For ongoing operations, the layout map helps operators and maintenance personnel navigate the facility and locate equipment or piping for repairs, upgrades, or inspections.
- Additionally, the layout map can include details for drainage and stormwater management, ensuring that rainwater is effectively channelled away from critical areas to prevent flooding.
- An example of a layout plan is given on the next page.

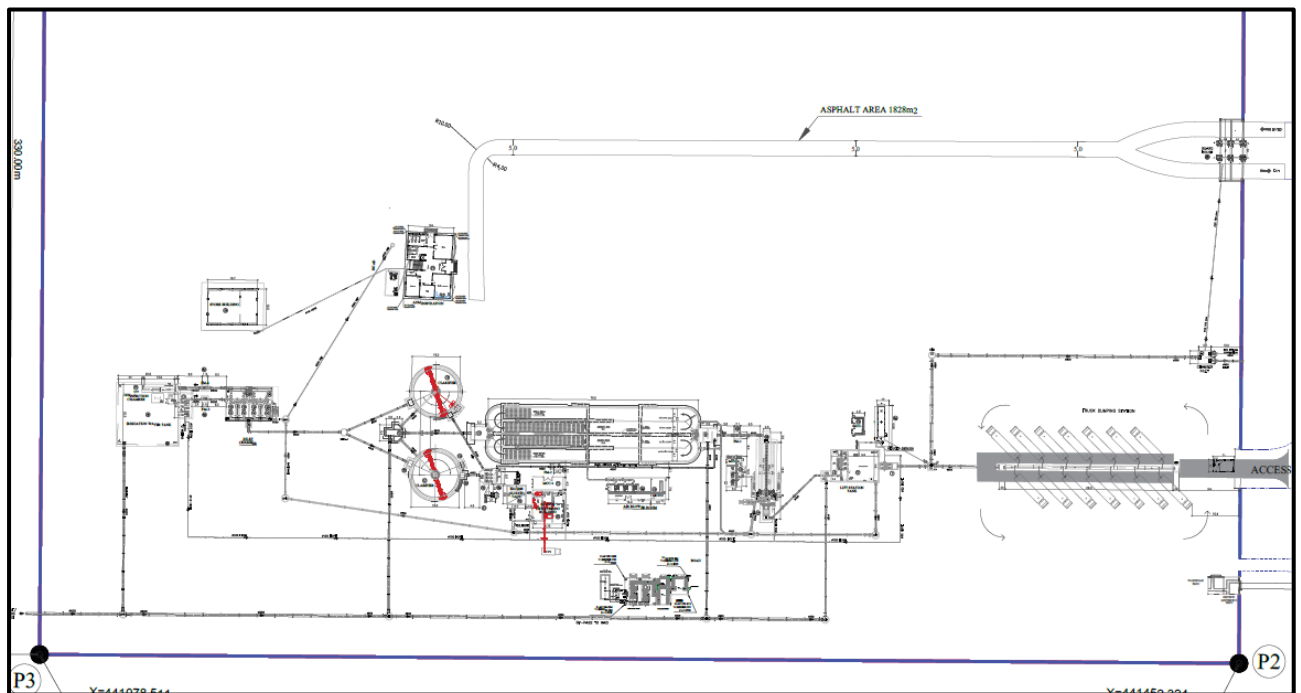
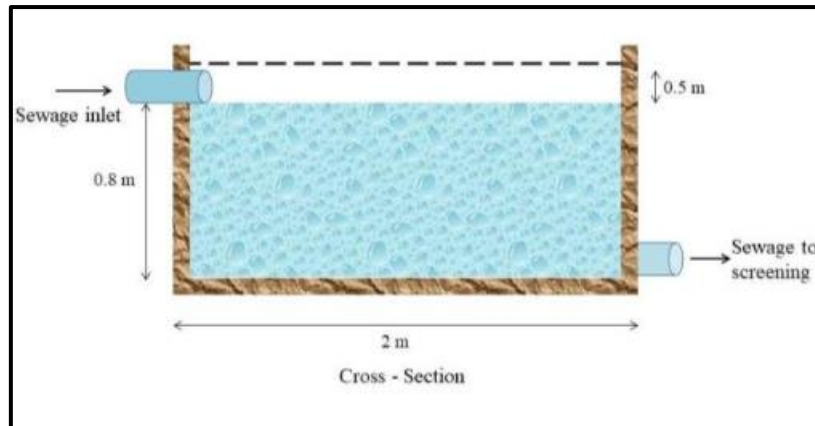


FIG: This image shows a layout for a WTP / STP plant.

- Note that the image provided may not be that clear since the original drawing made was on A1 sheet and this report is limited to A4, A3 sheets.

4 Equipment's used in WTP/STP in brief

4.1 Inlet Chamber



- The inlet chamber is where raw sewage or wastewater enters the treatment plant. It is the first stage of treatment where large debris is initially removed. This component helps regulate the flow and prevent large items from damaging downstream equipment.

4.2 Grit And Coarse Screen



- This section has mechanical screens and grit chambers that remove large particles, sand, and grit from the influent. Coarse screens typically remove larger debris, while finer screens remove smaller particles. Grit removal is essential to protect equipment and reduce wear.

4.3 Lift And Pump Station



- These pumps are used to lift the wastewater to higher levels as it moves through the plant, especially when gravity flow is not possible. They ensure smooth movement of water through various stages by providing the necessary pressure.

4.4 Air Blowers



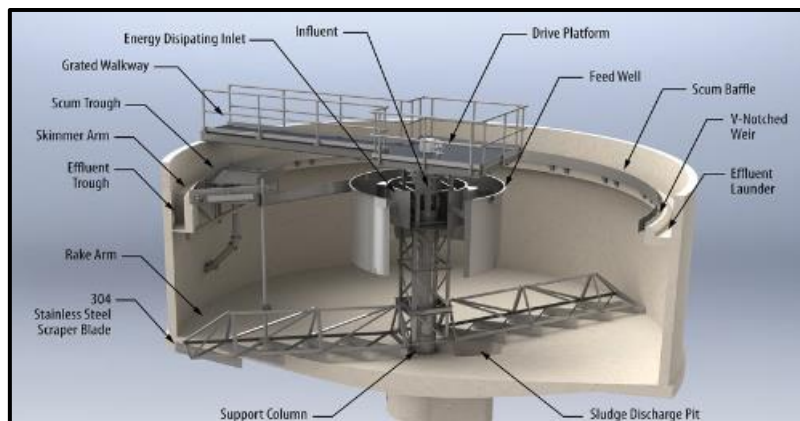
- Air blowers are used to provide aeration in various treatment tanks. Aeration is crucial for mixing and supporting biological treatment by supplying oxygen, which helps microorganisms break down organic matter.

4.5 Aeration Tanks



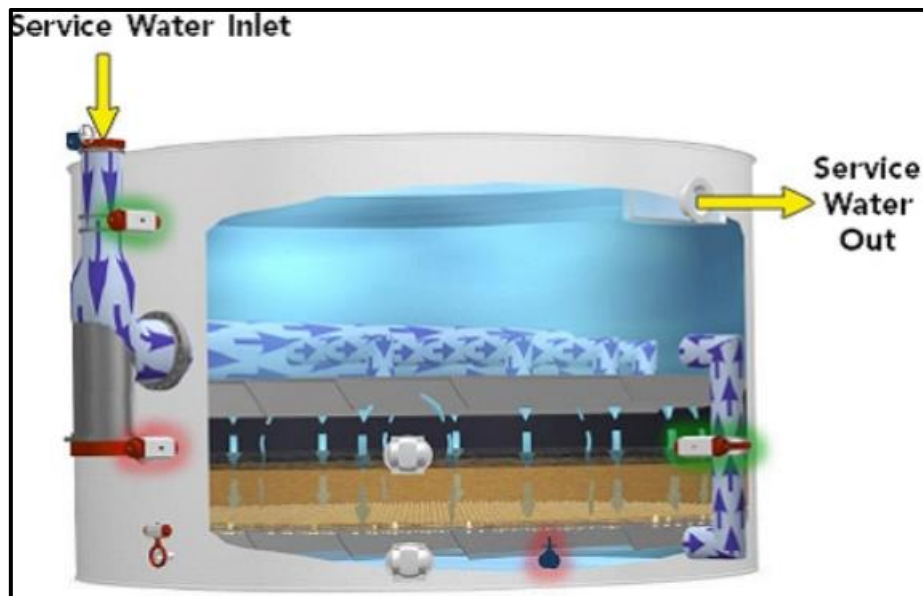
- The aeration tanks are central to biological treatment. In these tanks, microorganisms decompose organic pollutants in the presence of oxygen. This stage is vital for reducing the biological oxygen demand (BOD) in the wastewater.

4.6 Clarifier Tanks



- After aeration, the water flows into clarifiers, which are sedimentation tanks where the activated sludge settles at the bottom. The clarified effluent then moves to the next treatment stage. Clarifiers play an essential role in removing suspended solids.

4.7 Gravity Filters



- Gravity filters are used to filter out smaller particles remaining in the water after biological treatment. They generally use sand or other filtration media to trap and remove particulates, ensuring cleaner water.

4.8 Flowmeter and Monitoring Equipment's



- Flowmeters are distributed throughout the system to monitor the flow rate of water. Accurate flow measurement is crucial for controlling and optimizing the treatment process, ensuring compliance with treatment standards.

4.9 Activated Carbon or Odor Control Systems



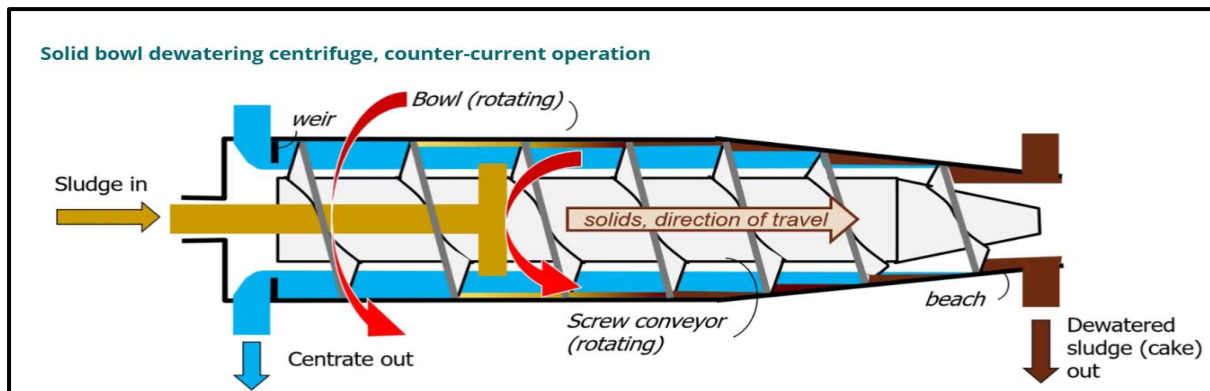
- This component uses activated carbon or other Odor-neutralizing systems to remove Odors from the wastewater before discharge or further treatment. It is especially useful in minimizing the release of unpleasant smells in populated areas.

4.10 Chlorination Chambers



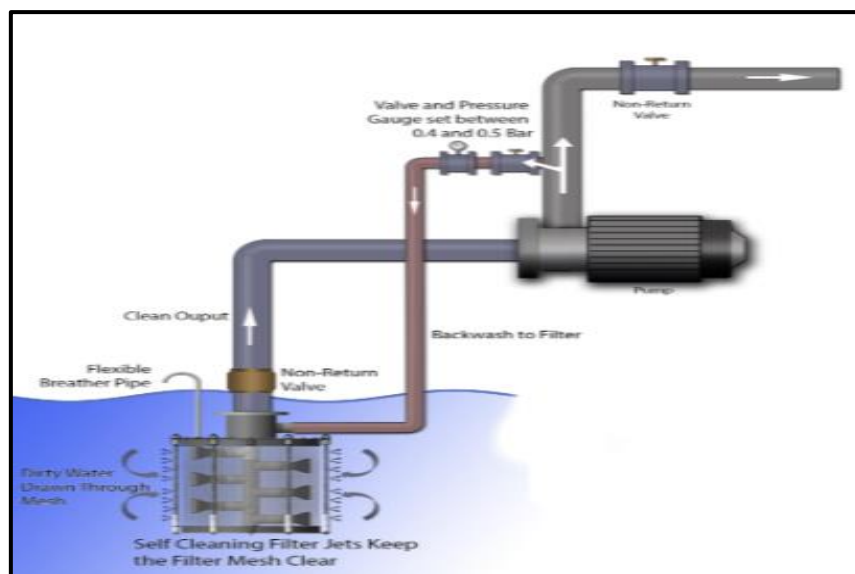
- The chlorination chamber is where disinfectants, typically chlorine, are added to kill remaining pathogens and ensure that the water is safe for discharge or reuse. Chlorination is a common final step in water treatment.

4.11 Sludge Handling Systems



- The sludge generated in clarifiers and other tanks is handled separately. The sludge may undergo further treatment, including thickening, dewatering, or digestion, before disposal or use as fertilizer. The sludge handling system ensures that waste by-products are managed responsibly.

4.12 Backwash Pump and Wash Water Tank



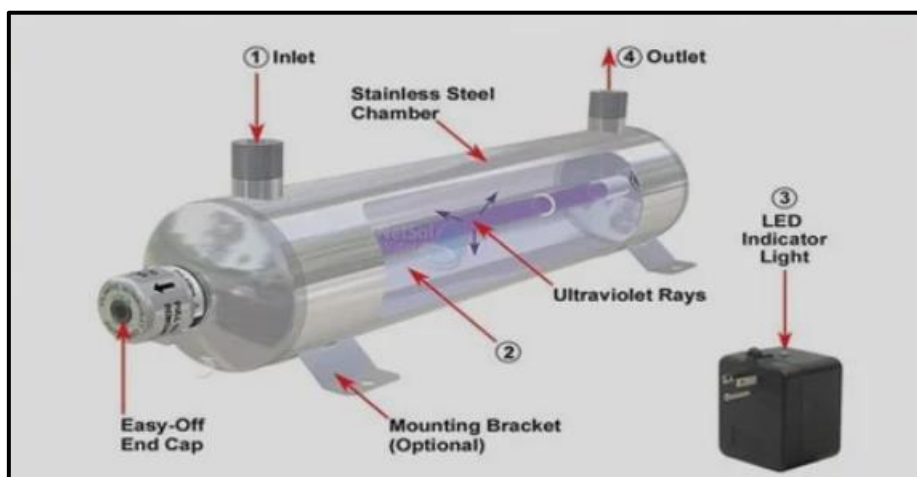
- Backwashing is a process used in filters to remove trapped particles. The backwash pump and tank ensure that filters are periodically cleaned, maintaining efficiency and preventing clogging. The wash water tank collects water for backwashing purposes.

4.13 Chemical Dosing Systems



- Various chemical dosing units are likely integrated at multiple points in the process. Chemicals may be added for coagulation, pH adjustment, or disinfection. Precise dosing is critical to achieve the desired water quality and maintain system balance.

4.14 UV Chambers



- UV Chambers are a critical component in water and wastewater treatment plants that use ultraviolet (UV) light to disinfect treated water before it's discharged or reused. In a UV chamber, the water flows through a compartment equipped with UV lamps that emit UV-C light, a wavelength particularly effective at destroying the DNA and RNA of harmful microorganisms, including bacteria, viruses, and parasites.

4.15 Submersible Pumps



- Submersible pumps are used to move water or wastewater from one part of the plant to another, often lifting it from lower to higher elevations where gravity flow isn't feasible. These pumps are designed to operate while submerged in the fluid, which makes them highly effective and efficient for treatment plant applications, as they do not require separate housing or protection from water exposure.

4.16 Return Sludge Pumps



- Return sludge pumps are specifically used to manage the sludge produced during the biological treatment processes in the wastewater treatment plant. Sludge is a mixture of solids and water, containing organic material and microorganisms that play a role in the treatment process.
- 0.75 factor is given that means that **75% of the incoming influent flow** (or the flow rate of the raw wastewater entering the treatment plant) is being returned as activated

sludge. This ratio helps control the concentration of microorganisms in the aeration tank, balancing the biological treatment process.

4.17 Anoxic Chamber



- An anoxic tank is a critical component in wastewater treatment plants, specifically in processes designed to remove nitrogen compounds from wastewater, such as denitrification. The term "anoxic" refers to conditions in the tank where oxygen is absent, but nitrate (NO_3^-) is present, which microorganisms use as an alternative electron acceptor instead of oxygen.

5 Some Design Calculations

5.1 Head Loss Calculation

- Head loss calculation is essential for Water Treatment Plants (WTP) and Sewage Treatment Plants (STP) because it determines the energy required to move water or wastewater through the system and ensures efficient design and operation of hydraulic components.

- Darcy's Weisbach equation is used for head loss calculations:

$$H_L = \frac{V^2}{2g} \left(\frac{fL}{D} + N_1K_1 + N_2K_2 + N_3K_3 + N_4K_4 \right)$$

- Colebrook equation:

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\varepsilon}{3.7D} + \frac{2.51}{R\sqrt{f}} \right)$$

- Renolds number calculation:

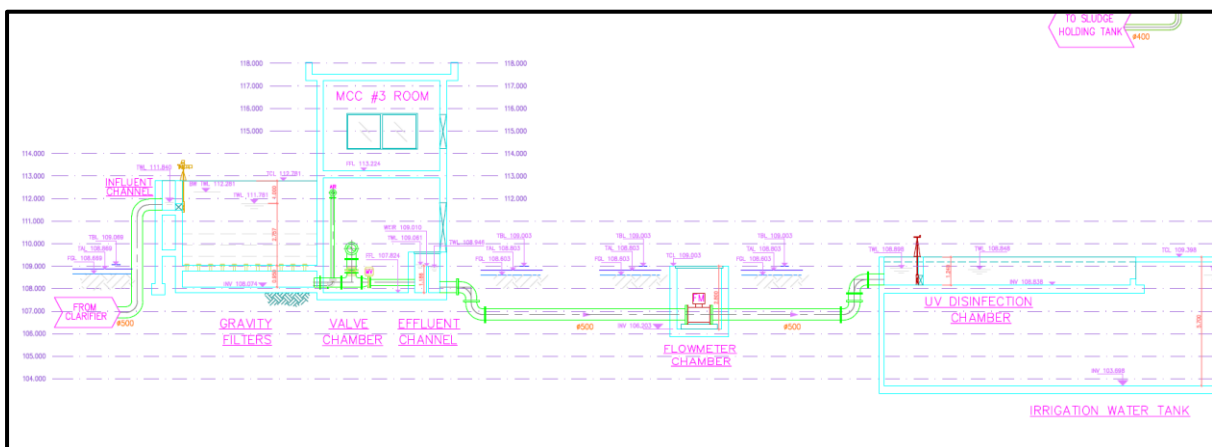
$$R = \frac{vD}{\mu}$$

TERMS:

- **HL:** Total head loss (in meters, m), which represents the energy loss due to friction and additional losses from fittings and bends.
- **V:** Flow velocity (in meters per second, m/s) of the fluid in the pipe.
- **g:** Acceleration due to gravity (9.81 m/s²).
- **F:** Friction factor, also known as the Darcy-Weisbach friction factor, which depends on the flow regime (laminar or turbulent) and the pipe's roughness.
- **L:** Length of the pipe (in meters, m), over which the fluid flows.
- **D:** Diameter of the pipe (in meters, m).
- **N₁:** Quantity of elbow 90 degree.
- **N₂:** Quantity of elbow 45 degree.
- **N₃:** Quantity of exits square.
- **N₄:** Quantity of entrance square.
- **K₁:** 0.3 for elbow 90 degree.
- **K₂:** 0.25 for elbow 45 degree.
- **K₃:** 1.0 for exits square.
- **K₄:** 0.5 for entrance square.
- **ε:** surface Roughness taken 1mm.
- **v:** Velocity.
- **μ:** Kinematic viscosity.

FOR EXAMPLE:

- For calculating the level of water from the UV chamber to gravity filter,



		design	peak	average	Units
Flow	Q	0.139	0.139	0.069	m ³ /s
Diameter	D	0.500	0.500	0.500	m
Radius	r	0.250	0.250	0.250	m
Area	a	0.196	0.196	0.196	m ²
Velocity	v	0.707	0.707	0.354	m/s
Reynolds number	Nr	3.51E+05	3.51E+05	1.76E+05	
Roughness of pipe	e	1.000	1.000	1.000	mm
friction coeff	f	0.024	0.024	0.024	trial
Cole-Brook Eqn	f	0.024	0.024	0.024	
Pipe length	L	16	16	16	m
Elbows 90 deg	n1	2	2	2	no.
Elbows 45 deg	n2	0	0	0	no.
Exits square	n3	0	0	0	no.
Entrance square	n4	1	1	1	no.
Headloss	H _t	0.048	0.048	0.012	m

- The conditions occurring during the peak flow is considered for designing.
- The top water level at the UV chamber is 108.898 meters.
- The head loss calculated is 0.048 meters.
- Therefore, the top water level at the gravity filter is:

$$108.898 + 0.048 = \underline{108.946 \text{ meters}}$$

- Python program supporting the calculations:

```
import sympy as sp
import math
import pandas as pd

# User input parameters
Average_Flow1 = float(input("AVERAGE FLOW (m3/day): "))
Average_Flow2 = Average_Flow1 / (24 * 60 * 60)

Peak_Flow1 = float(input("PEAK FLOW (m3/day): "))
Peak_Flow2 = Peak_Flow1 / (24 * 60 * 60)

Diameter = float(input("DIAMETER (m): "))
Radius = Diameter / 2
Area = math.pi * (Radius ** 2)

# Velocity
Peak_Velocity = Peak_Flow2 / Area
Average_Velocity = Average_Flow2 / Area

# Kinematic viscosity
Kinematic_Viscosity = float(input("KINEMATIC VISCOSITY (m2/s): "))

# Reynolds Number
Peak_Reynolds_Number = (Peak_Velocity * Diameter) / Kinematic_Viscosity
Average_Reynolds_Number = (Average_Velocity * Diameter) / Kinematic_Viscosity

# Roughness and pipe length
ROUGHNESS_OF_PIPE = float(input("ROUGHNESS OF PIPE (m): "))
Length_Of_Pipe = float(input("LENGTH OF PIPE (m): "))

# Pipe components
Number_Of_90_Degree_Elbows = int(input("NUMBER OF 90 DEGREE ELBOWS: "))
Number_Of_45_Degree_Elbows = int(input("NUMBER OF 45 DEGREE ELBOWS: "))
Number_Of_Exit_Square = int(input("NUMBER OF EXIT SQUARES: "))
Number_Of_ENTRANCE_Square = int(input("NUMBER OF ENTRANCE SQUARES: "))

# Solve Colebrook equation for friction factor
friction_factor = 0.024 # Get numerical value of friction factor
```

```

# Head loss calculation
Head_Loss_Peak = ((Peak_Velocity ** 2) / (2 * 9.81)) * (((friction_factor * Length_Of_Pipe) / Diameter) +
(0.3 * Number_Of_90_Degree_Elbows) +
(0.25 * Number_Of_45_Degree_Elbows) +
(1 * Number_Of_Exit_Square) +
(0.5 * Number_Of_ENTRANCE_Square))

Head_Loss_Average = ((Average_Velocity ** 2) / (2 * 9.81)) * (((friction_factor * Length_Of_Pipe) / Diameter) +
(0.3 * Number_Of_90_Degree_Elbows) +
(0.25 * Number_Of_45_Degree_Elbows) +
(1 * Number_Of_Exit_Square) +
(0.5 * Number_Of_ENTRANCE_Square))

# Data to display in table
SrNo = list(range(1, 14))
Parameters = [
    "FLOW", "DIAMETER", "RADIUS", "VELOCITY", "REYNOLDS NUMBER",
    "ROUGHNESS OF PIPE", "LENGTH OF PIPE", "NUMBER OF 90 DEGREE ELBOWS",
    "NUMBER OF 45 DEGREE ELBOWS", "NUMBER OF EXIT SQUARES", "NUMBER OF ENTRANCE SQUARES",
    "FRICTION FACTOR", "HEAD LOSS"
]

Peak = [
    f"{Peak_Flow2:.3f}", f"{Diameter:.3f}", f"{Radius:.3f}", f"{Peak_Velocity:.3f}",
    f"{Peak_Reynolds_Number:.3f}", f"{ROUGHNESS_OF_PIPE:.3f}", f"{Length_Of_Pipe:.3f}",
    f"{Number_Of_90_Degree_Elbows}", f"{Number_Of_45_Degree_Elbows}", f"{Number_Of_Exit_Square}",
    f"{Number_Of_ENTRANCE_Square}", f"{friction_factor:.3f}", f"{Head_Loss_Peak:.3f}"
]

Average = [
    f"{Average_Flow2:.3f}", f"{Diameter:.3f}", f"{Radius:.3f}", f"{Average_Velocity:.3f}",
    f"{Average_Reynolds_Number:.3f}", f"{ROUGHNESS_OF_PIPE:.3f}", f"{Length_Of_Pipe:.3f}",
    f"{Number_Of_90_Degree_Elbows}", f"{Number_Of_45_Degree_Elbows}", f"{Number_Of_Exit_Square}",
    f"{Number_Of_ENTRANCE_Square}", f"{friction_factor:.3f}", f"{Head_Loss_Average:.3f}"
]

units = [
    "m3/day", "m", "m", "m/s", "unitless", "m", "m", "qty", "qty", "qty", "qty",
    "unitless", "m"
]

# Create the DataFrame
table_data = {"SrNo": SrNo, "Parameters": Parameters, "Peak": Peak, "Average": Average, "Units": units}
table = pd.DataFrame(table_data)

# Display the table
print(table)

```

➤ Results:

```

AVERAGE FLOW (m3/day): 6000
PEAK FLOW (m3/day): 12000
DIAMETER (m): 0.5
KINEMATIC VISCOSITY (m2/s): 1.007e-6
ROUGHNESS OF PIPE (m): 0.001
LENGTH OF PIPE (m): 16
NUMBER OF 90 DEGREE ELBOWS: 2
NUMBER OF 45 DEGREE ELBOWS: 0
NUMBER OF EXIT SQUARES: 0
NUMBER OF ENTRANCE SQUARES: 1

```

	SrNo	Parameters	Peak	Average	Units
0	1	FLOW	0.139	0.069	m3/day
1	2	DIAMETER	0.500	0.500	m
2	3	RADIUS	0.250	0.250	m
3	4	VELOCITY	0.707	0.354	m/s
4	5	REYNOLDS NUMBER	351219.117	175609.559	unitless
5	6	ROUGHNESS OF PIPE	0.001	0.001	m
6	7	LENGTH OF PIPE	16.000	16.000	m
7	8	NUMBER OF 90 DEGREE ELBOWS	2	2	qty
8	9	NUMBER OF 45 DEGREE ELBOWS	0	0	qty
9	10	NUMBER OF EXIT SQUARES	0	0	qty
10	11	NUMBER OF ENTRANCE SQUARES	1	1	qty
11	12	FRICTION FACTOR	0.024	0.024	unitless
12	13	HEAD LOSS	0.048	0.012	m

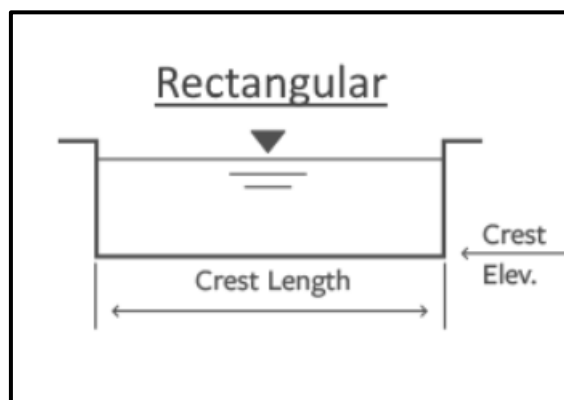
5.2 Wier And its Calculation

- A **weir** is a barrier built across an open water channel, such as a river or a canal, designed to control, measure, or divert water flow. In the context of water and sewage treatment plants, weirs play a critical role in managing water levels and flow rates within treatment basins or tanks. The concept is similar to a dam but typically on a smaller scale, and weirs are usually designed to allow controlled overflow.
- Even though there are many types of weirs, mostly rectangular and triangular weirs were used that I came across.

Types of Weirs:

Rectangular Weir:

- **Description:** A rectangular weir has a horizontal, straight crest, which forms a rectangular opening for water to flow over.



- Formula:

$$Q = 1.84 LH^{1.5}$$

Q = flow rate.

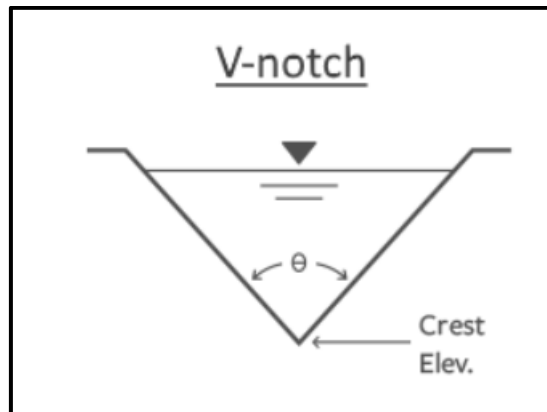
H = head.

L = Length of Weir.

- **Function and Use:** Rectangular weirs are among the simplest and most common weir types, often used in water treatment plants, rivers, and canals. They are well-suited for measuring flow over a wide range of rates.

V-Notch Weir:

- **Description:** The V-notch weir has a triangular opening, typically with a 90-degree angle at the base, though other angles can also be used.



- **Formula:**

$$Q = 1.38 H^{2.5}$$

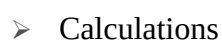
Q = flow rate.

H = head.

- **Function and Use:** V-notch weirs are highly sensitive to low flows, which makes them ideal for precise flow measurements in low-flow conditions. They are often used in laboratory setups, small streams, and areas where exact flow measurement is critical.

For example:

- For calculating head over weir from clarifier to launder.
- If the invert level of weir is 112.224 meters.
- There for calculating head over weir for a v notch type weir, we will have to first divide the flow rate by number of weirs.

32 | Page

➤ Python program supporting the calculations:

```
import pandas as pd
# Input weir type
print("TYPE: 'TRIANGULAR' OR 'RECTANGULAR'")
weir_type = input("TYPE OF WEIR: ").strip().upper()
# Check the weir type and perform calculations
if weir_type == "TRIANGULAR":
    # Triangular weir calculations
    flow_rate = float(input("FLOW RATE (m^3/s): "))
    number_of_weirs = int(input("NUMBER OF WEIRS: "))
    constant = 1.38
    flow_per_weir = flow_rate / number_of_weirs
    head_over_weir = (flow_per_weir / constant) ** (1 / 2.5)
    # Create table
    data = {
        "SrNo": [1, 2, 3, 4],
        "Parameters": ["FLOW RATE", "NUMBER OF WEIR", "FLOW PER WEIR", "HEAD OVER WEIR"],
        "Design": [f"{flow_rate:.3f}", number_of_weirs, f"{flow_per_weir:.3f}", f"{head_over_weir:.3f}"],
        "Units": ["m³/s", "qty", "m³/s", "m"]
    }
elif weir_type == "RECTANGULAR":
    # Rectangular weir calculations
    flow_rate = float(input("FLOW RATE (m^3/s): "))
    number_of_weirs = int(input("NUMBER OF WEIRS: "))
    length = float(input("LENGTH OF WEIR (m): "))
    constant = 1.84
    flow_per_weir = flow_rate / number_of_weirs
    head_over_weir = (flow_per_weir / (constant * length)) ** (1 / 1.5)
    # Create table
    data = {
        "SrNo": [1, 2, 3, 4, 5],
        "Parameters": ["FLOW RATE", "LENGTH OF WEIR", "NUMBER OF WEIR", "FLOW PER WEIR", "HEAD OVER WEIR"],
        "Design": [f"{flow_rate:.3f}", f"{length:.3f}", number_of_weirs, f"{flow_per_weir:.3f}", f"{head_over_weir:.3f}"],
        "Units": ["m³/s", "m", "qty", "m³/s", "m"]
    }
else:
    print("ERROR")
    data = {}
# Display the table if data is available
if data:
    table = pd.DataFrame(data)
    print(table)
```

➤ Results:

```
TYPE: 'TRIANGULAR' OR 'RECTANGULAR'
TYPE OF WEIR: TRIANGULAR
FLOW RATE (m^3/s): 0.139
NUMBER OF WEIRS: 322
```

	SrNo	Parameters	Design	Units
0	1	FLOW RATE	0.139	m³/s
1	2	NUMBER OF WEIR	322	qty
2	3	FLOW PER WEIR	0.000	m³/s
3	4	HEAD OVER WEIR	0.040	m

➤ Therefore,

$$112.224 + 0.040 = \underline{112.264 \text{ meters}}$$

5.3 Pipe Sizing Calculation

- Pipe sizing is a critical aspect of designing water and wastewater treatment systems. Properly sized pipes ensure efficient fluid transport, minimize energy costs, and prevent issues such as cavitation, excessive pressure drops, and insufficient flow. Sizing a pipe involves determining the right diameter and thickness based on flow requirements, fluid properties, pressure constraints, and system layout
- Pipe sizing is done for every pipe that carry a fluid within them, weather it is from gravity, pumps or air blowers.
- In pipe sizing calculations the pipes that has a fluid transferring through gravity has a larger diameter than the pipe that acts under the action of pumps.
- Usually, the velocity range in gravity pipes is between 0.4 to 0.9 m/s.
- And the velocity range for pump acting pipes is between 1 – 2 m/s

For Example,

SrNo.	Parameters	Design	Units
1	Flow Rate through per pump:	0.069444	m ³ /s
2	Number of pumps in use:	2	qty
3	Flow rate through a pipe:	0.138889	
4	Velocity:	1.15	m/s
5	Diameter:	0.392238	m
6	Rounded Diameter to the nearest pipe size:	0.4	m
SrNo.	Parameters	Design	Units
1	Flow Rate through per pump:	0.086806	m ³ /s
2	Number of pumps in use:	5	qty
3	Flow rate through a pipe:	0.434028	
4	Velocity:	1.6	m/s
5	Diameter:	0.587847	m
6	Rounded Diameter to the nearest pipe size:	0.6	m

- Pipe sizing calculation are done for a pump in parallel, where the flowrate is combined.

For AIR BLOWERS,

For Example,

SrNo.	Parameters:	Design:	Units:	
1	Flow rate per Blower:	0.833333	m ³ /s	
2	Number of blower in use at a time:	2	qty	
3	Flow rate:	1.666667	m ³ /s	
4	Velocity:	15	m/s	
5	Diameter:	0.376222	m	
6	Rounded Diameter:	0.4	m	
SrNo.	Parameters:	Design:	Units:	
1	Flow rate per Blower:	0.969444	m ³ /s	
2	Number of blower in use at a time:	1	qty	
3	Flow rate:	0.969444	m ³ /s	
4	Velocity:	15	m/s	
5	Diameter:	0.286933	m	
6	Rounded Diameter:	0.3	m	
SrNo.	Parameters:	Design:	Units:	
1	Flow rate per Blower:	1.25	m ³ /s	
2	Number of blower in use at a time:	2	qty	
3	Flow rate:	2.5	m ³ /s	
4	Velocity:	15	m/s	
5	Diameter:	0.460776	m	
6	Rounded Diameter:	0.5	m	

- PVC, Stainless steel or HDPE such types of pipes are generally used.
- Pressurized pipes need to withstand high pressure and occasional surges, so they're designed with thicker walls and strong joints. Materials like HDPE are often used for their flexibility and corrosion resistance, which is beneficial when carrying water with potential chemicals or pollutants.

Gravity pipes are installed at a slope, relying on gravitational flow rather than pressure. They need to resist corrosion, abrasion, and potential blockages, especially in waste applications.

Air blower pipes need to withstand moderate pressure and temperature changes while being corrosion-resistant to avoid contamination.

6 Project Management Through Primavera P6

6.1 Key Features of PRIMAVERA P6

➤ Project Scheduling and Planning

- Primavera P6 allows users to create detailed project schedules with precise timelines. Using a Gantt chart interface, users can define project activities, assign durations, and organize them into work breakdown structures (WBS) that help break down complex projects into manageable tasks.
- The software supports multiple project views and scheduling methods, including critical path method (CPM) and resource levelling, making it possible to view and optimize timelines.
- Customization of a calendar which determines the length of a project can also be done.

➤ Resource and Cost Management

- Primavera P6 offers detailed resource management tools that help project managers allocate, track, and optimize resources (e.g., labour, materials, equipment) across different activities and projects.
- Cost management features allow users to track actual and forecasted costs, integrate with financial systems, and manage project budgets effectively. The software enables control over expenditures, contributing to better decision-making and budget adherence.

➤ Risk Management

- Primavera P6 includes risk analysis tools that allow users to identify potential risks and uncertainties, analyse their impact on project timelines and resources, and prepare mitigation strategies.
- The software helps with developing contingency plans and can calculate risk-adjusted schedules to ensure realistic expectations and proactive risk management.

➤ Progress Tracking and Reporting

- Users can monitor real-time project progress and performance using visual dashboards and custom reports. Primavera P6 generates various progress reports, including earned value management (EVM) reports, which allow tracking of project costs, schedule adherence, and other KPIs
- Primavera P6's reporting tools facilitate easy communication of project status to stakeholders, helping ensure alignment and accountability.

➤ Collaboration and Communication

- Primavera P6 provides collaborative features that allow multiple team members to update, view, and interact with the project plan, enhancing communication within teams and with stakeholders.
- The software can integrate with other tools (e.g., Excel, ERP systems) to streamline workflows and ensure all team members have access to current project information.

➤ Multi-Project and Multi-User Management

- Primavera P6 is designed to manage multiple projects simultaneously, making it ideal for large organizations handling several ongoing projects.
- It supports multi-user access, allowing users to work on various project elements concurrently while maintaining project security and control over data.

➤ Forecasting and Scenario Analysis

- Primavera P6 allows users to create ****what-if scenarios**** to simulate different project conditions or changes, such as resource constraints, project delays, or budget cuts.
- Users can analyse the impact of these scenarios on the project's timeline and budget, aiding in better decision-making and strategic planning.

6.2 Gantt chart and Activity Lists

In Primavera P6, a Gantt chart and an **activity list** are essential tools for project planning and scheduling, helping project managers and stakeholders visualize project progress and manage tasks effectively.

Gantt Chart

- Gantt chart in Primavera P6 is a graphical representation of a project's schedule, where activities are displayed as horizontal bars across a timeline. The length and position of each bar represent the start date, duration, and end date of an activity. Primavera's Gantt chart is highly interactive, showing dependencies between activities, critical paths, and resource assignments.
- Activities: Each horizontal bar on the Gantt chart corresponds to a project activity, giving a visual overview of its duration.
- Timeline: The x-axis represents the timeline (typically days, weeks, or months), while the y-axis lists project activities.
- Dependencies: Lines between bars indicate dependencies, showing relationships, such as Finish-to-Start or Start-to-Start dependencies between activities.
- Critical Path: Critical tasks, which are vital for the timely completion of the project, are often highlighted, helping project managers focus on key activities.

What the Gantt Chart Shows:

- The sequence of activities and their timelines.
- Start and end dates for each activity.
- Milestones (key project events) for monitoring project progress.
- The project's overall progress compared to planned timelines.

Activity List

- The activity list in Primavera P6 is a detailed, sequential listing of all project tasks or activities, breaking down the project into smaller, manageable components. Each activity in this list is a specific task that must be completed for the project to move forward. In Primavera P6, activities are grouped under Work Breakdown Structure (WBS) categories, which organize tasks hierarchically according to project phases or areas.

Each entry in the activity list typically includes:

- Activity ID and Name: A unique identifier and descriptive title for the task.
- Duration: The estimated time needed to complete the activity.
- Start and Finish Dates: Scheduled dates for the beginning and end of each activity.
- Predecessors and Successors: Dependencies showing which activities must occur before or after a given activity.
- Resources: Allocation of labour, equipment, or materials needed for the activity.

What the Activity List Shows:

- The breakdown of tasks and subtasks in the project.
 - Dependencies and sequence of tasks, helping prioritize activities.
 - Duration and schedule details for each activity.
 - Assignment of resources, helping optimize resource usage.
-
- In Primavera P6, the Gantt chart and activity list work together to provide a comprehensive overview of the project, allowing managers to track progress, adjust schedules, allocate resources, and meet project deadlines.

These terms are commonly used in primavera p6 necessary for understanding the charts

- **Baseline:**
Baseline data refers to the approved version of a project plan, including the schedule, costs, and scope. This data serves as a standard for measuring project performance.
- **BL Project Start:**
This is the date when the project officially begins. It marks the start of project activities and the execution of the project plan.
- **BL Project Finish:**
This is the date when the project is expected to be completed. It signifies the end of project activities, ideally aligned with the project's goals and deliverables.

➤ **Budgeted Total Cost:**

This is the total budget allocated for the project. It includes all planned expenses required to complete the project, often derived from the project plan.

➤ **Actual Total Cost**

(ATC: This is the total cost incurred so far in the project, reflecting all expenses that have been paid or billed. This figure is important for monitoring budget adherence.

➤ **Planned Value (PV):**

Also known as Budgeted Cost of Work Scheduled (BCWS), this represents the value of the work that was planned to be completed by a specific time in the project. It's calculated based on the project's schedule and budget.

➤ **Cost Earned Value (EV):**

Also referred to as Earned Value (EV), this indicates the value of the work actually completed by a specific point in time. It is calculated by multiplying the percentage of work completed by the total budget.

➤ **Cost Performance Index (CPI):**

This is a measure of cost efficiency in the project. It is calculated using the form

$$CPI = \frac{EV}{ATC}$$

A CPI greater than 1 indicates that the project is under budget, while a CPI less than 1 indicates it is over budget.

➤ **Performance % Complete:**

This metric shows the percentage of work that has been completed relative to the total project work. It can be calculated based on deliverables, tasks, or phases.

➤ **Schedule % Complete:**

This represents the percentage of the project schedule that has been completed. It compares the actual progress of the project timeline against the planned schedule.

➤ **Schedule performance index:**

is a key performance metric used in project management to assess the efficiency of a project in terms of its schedule. It is a part of the Earned Value Management (EVM) methodology, which integrates project scope, schedule, and cost to provide a comprehensive view of project performance.

$$SPI = \frac{EV}{PV}$$

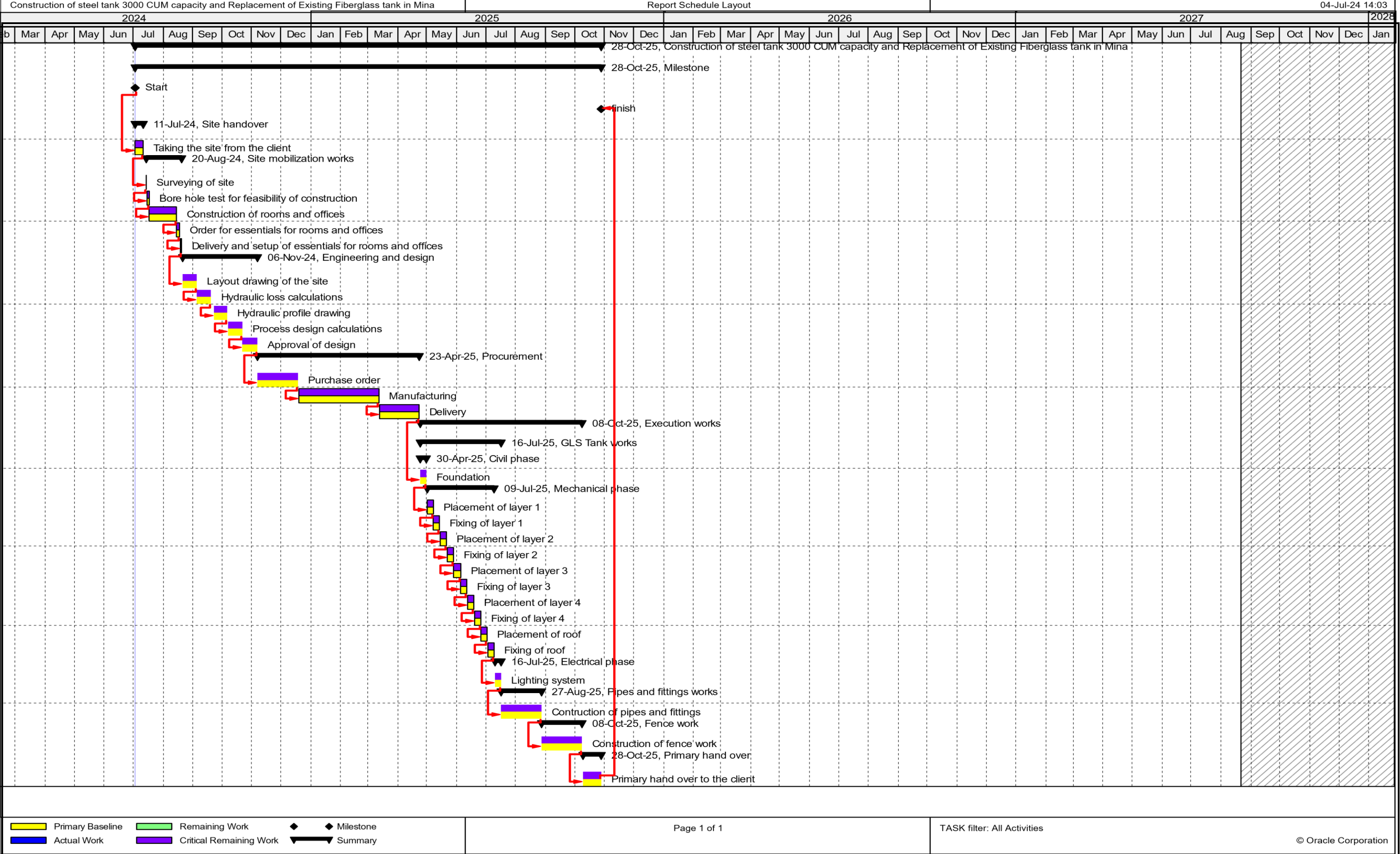
A SPI greater than 1 indicates that more work is done than what planned, while a SPI less than 1 indicates less work is done than planned.

➤ **Data date:**

is a key reference point in project scheduling and tracking. It represents the point in time at which the schedule is considered current.

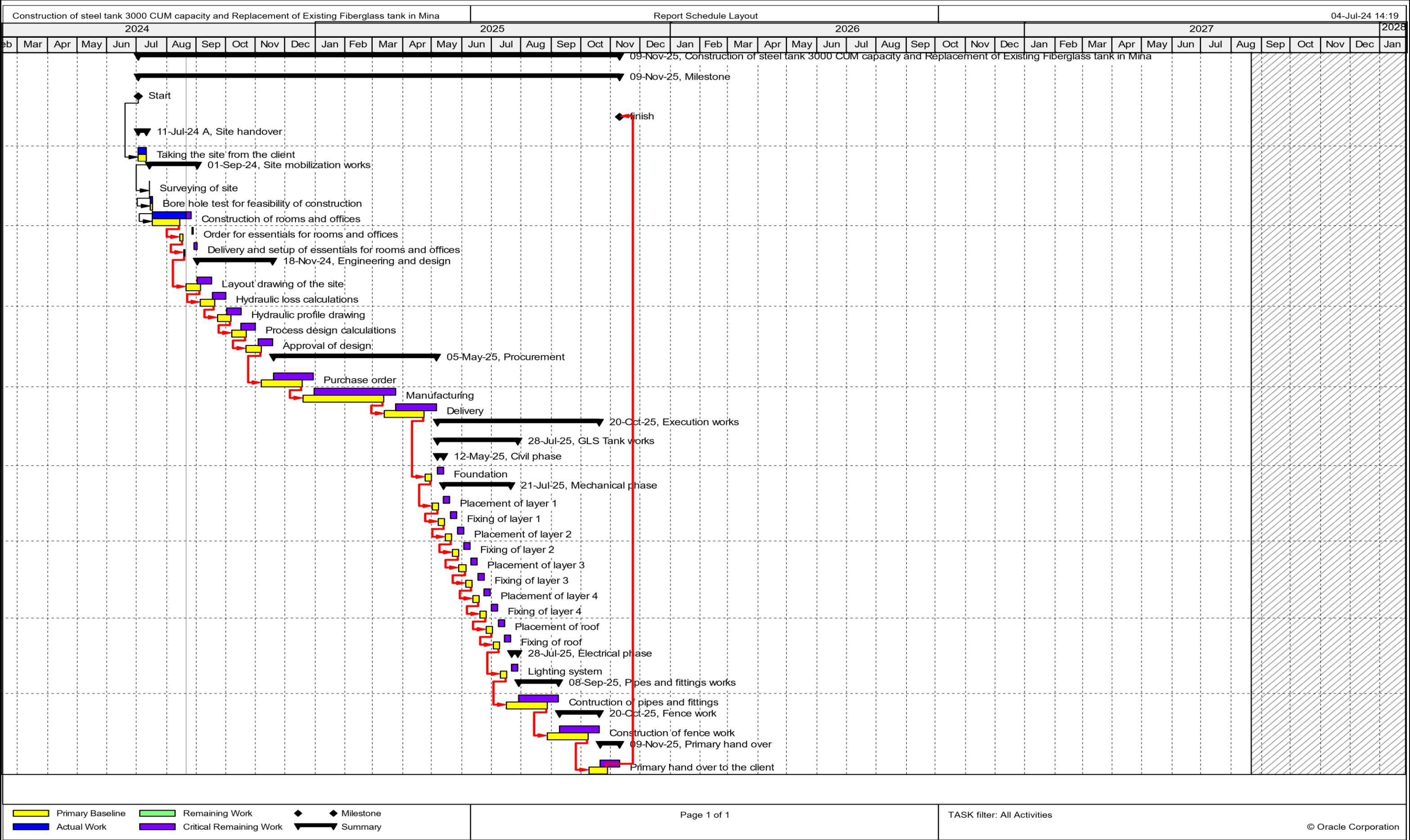
- It is important to note that during my internship, I was assigned tasks related to their previous projects. As a result, my task list was relatively small, and my critical path did not have many sub-branches and making a proper schedule needed myself to be in contact with the client which wasn't possible for an intern who has a year left for degree completion.
- Additionally, I can include one of my completed works in my report, as the extensive activity list requires A1 and A2 sheets to be displayed clearly. And this report is made on A4 sheets and few A3 sheets.
- Task related to steel tank installation was given: which is shown on next pages.
- The data date was supposed to be given at 4 instances

Construction of steel tank 3000 CUM capacity and Replacement of Existing Fiberglass tank in Mina					Report Schedule Layout					04-Jul-24 13:50			
Activity ID	Activity Name	Original Duration	Start	Finish	BL Project Start	BL Project Finish	Budgeted Total Cost	Actual Total Cost	Planned Value Cost	Earned Value Cost	Performance % Complete	Schedule % Complete	
 Construction of steel tank 3000 CUM c		345	03-Jul-24	28-Oct-25	03-Jul-24	28-Oct-25	52,977,531.70	0.00	0.00	0.00	0%	0%	
 Milestone		345	03-Jul-24	28-Oct-25	03-Jul-24	28-Oct-25	0.00	0.00	0.00	0.00	0%	0%	
 A	Start	0	03-Jul-24		03-Jul-24		0.00	0.00	0.00	0.00	0%	0%	
 B	finish	0		28-Oct-25		28-Oct-25	0.00	0.00	0.00	0.00	0%	0%	
 Site handover		7	03-Jul-24	11-Jul-24	03-Jul-24	11-Jul-24	0.00	0.00	0.00	0.00	0%	0%	
 1	Taking the site from the client	7	03-Jul-24	11-Jul-24	03-Jul-24	11-Jul-24	0.00	0.00	0.00	0.00	0%	0%	
 Site mobilization works		28	14-Jul-24	20-Aug-24	14-Jul-24	20-Aug-24	0.00	0.00	0.00	0.00	0%	0%	
 2	Surveying of site	1	14-Jul-24	14-Jul-24	14-Jul-24	14-Jul-24	0.00	0.00	0.00	0.00	0%	0%	
 3	Bore hole test for feasibility of construction	3	15-Jul-24	17-Jul-24	15-Jul-24	17-Jul-24	0.00	0.00	0.00	0.00	0%	0%	
 4	Construction of rooms and offices	20	18-Jul-24	14-Aug-24	18-Jul-24	14-Aug-24	0.00	0.00	0.00	0.00	0%	0%	
 5	Order for essentials for rooms and offices	2	15-Aug-24	18-Aug-24	15-Aug-24	18-Aug-24	0.00	0.00	0.00	0.00	0%	0%	
 6	Delivery and setup of essentials for rooms and office	2	19-Aug-24	20-Aug-24	19-Aug-24	20-Aug-24	0.00	0.00	0.00	0.00	0%	0%	
 Engineering and design		56	21-Aug-24	06-Nov-24	21-Aug-24	06-Nov-24	21,860.80	0.00	0.00	0.00	0%	0%	
 7	Layout drawing of the site	11	21-Aug-24	04-Sep-24	21-Aug-24	04-Sep-24	6,090.00	0.00	0.00	0.00	0%	0%	
 8	Hydraulic loss calculations	11	05-Sep-24	19-Sep-24	05-Sep-24	19-Sep-24	1,200.00	0.00	0.00	0.00	0%	0%	
 9	Hydraulic profile drawing	11	22-Sep-24	06-Oct-24	22-Sep-24	06-Oct-24	1,556.80	0.00	0.00	0.00	0%	0%	
 10	Process design calculations	11	07-Oct-24	21-Oct-24	07-Oct-24	21-Oct-24	7,014.00	0.00	0.00	0.00	0%	0%	
 11	Approval of design	12	22-Oct-24	06-Nov-24	22-Oct-24	06-Nov-24	6,000.00	0.00	0.00	0.00	0%	0%	
 Procurement		120	07-Nov-24	23-Apr-25	07-Nov-24	23-Apr-25	12,547,345.00	0.00	0.00	0.00	0%	0%	
 12	Purchase order	30	07-Nov-24	18-Dec-24	07-Nov-24	18-Dec-24	2,047,345.00	0.00	0.00	0.00	0%	0%	
 13	Manufacturing	60	19-Dec-24	12-Mar-25	19-Dec-24	12-Mar-25	6,000,000.00	0.00	0.00	0.00	0%	0%	
 14	Delivery	30	13-Mar-25	23-Apr-25	13-Mar-25	23-Apr-25	4,500,000.00	0.00	0.00	0.00	0%	0%	
 Execution works		120	24-Apr-25	08-Oct-25	24-Apr-25	08-Oct-25	40,268,276.90	0.00	0.00	0.00	0%	0%	
 GLS Tank works		60	24-Apr-25	16-Jul-25	24-Apr-25	16-Jul-25	38,385,727.00	0.00	0.00	0.00	0%	0%	
 Civil phase		5	24-Apr-25	30-Apr-25	24-Apr-25	30-Apr-25	2,135,727.00	0.00	0.00	0.00	0%	0%	
 15	Foundation	5	24-Apr-25	30-Apr-25	24-Apr-25	30-Apr-25	2,135,727.00	0.00	0.00	0.00	0%	0%	
 Mechanical phase		50	01-May-25	09-Jul-25	01-May-25	09-Jul-25	33,750,000.00	0.00	0.00	0.00	0%	0%	
 16	Placement of layer 1	5	01-May-25*	07-May-25	01-May-25	07-May-25	6,250,000.00	0.00	0.00	0.00	0%	0%	
 17	Fixing of layer 1	5	08-May-25	14-May-25	08-May-25	14-May-25	500,000.00	0.00	0.00	0.00	0%	0%	
 18	Placement of layer 2	5	15-May-25	21-May-25	15-May-25	21-May-25	6,250,000.00	0.00	0.00	0.00	0%	0%	
 19	Fixing of layer 2	5	22-May-25	28-May-25	22-May-25	28-May-25	500,000.00	0.00	0.00	0.00	0%	0%	
 20	Placement of layer 3	5	29-May-25	04-Jun-25	29-May-25	04-Jun-25	6,250,000.00	0.00	0.00	0.00	0%	0%	
 21	Fixing of layer 3	5	05-Jun-25	11-Jun-25	05-Jun-25	11-Jun-25	500,000.00	0.00	0.00	0.00	0%	0%	
 22	Placement of layer 4	5	12-Jun-25	18-Jun-25	12-Jun-25	18-Jun-25	6,250,000.00	0.00	0.00	0.00	0%	0%	
 23	Fixing of layer 4	5	19-Jun-25	25-Jun-25	19-Jun-25	25-Jun-25	500,000.00	0.00	0.00	0.00	0%	0%	
 24	Placement of roof	5	26-Jun-25	02-Jul-25	26-Jun-25	02-Jul-25	6,250,000.00	0.00	0.00	0.00	0%	0%	
 25	Fixing of roof	5	03-Jul-25	09-Jul-25	03-Jul-25	09-Jul-25	500,000.00	0.00	0.00	0.00	0%	0%	
 Electrical phase		5	10-Jul-25	16-Jul-25	10-Jul-25	16-Jul-25	2,500,000.00	0.00	0.00	0.00	0%	0%	
 26	Lighting system	5	10-Jul-25	16-Jul-25	10-Jul-25	16-Jul-25	2,500,000.00	0.00	0.00	0.00	0%	0%	
 Pipes and fittings works		30	17-Jul-25	27-Aug-25	17-Jul-25	27-Aug-25	1,528,499.50	0.00	0.00	0.00	0%	0%	
 27	Contruction of pipes and fittings	30	17-Jul-25	27-Aug-25	17-Jul-25	27-Aug-25	1,528,499.50	0.00	0.00	0.00	0%	0%	
 Fence work		30	28-Aug-25	08-Oct-25	28-Aug-25	08-Oct-25	354,050.40	0.00	0.00	0.00	0%	0%	
 28	Construction of fence work	30	28-Aug-25	08-Oct-25	28-Aug-25	08-Oct-25	354,050.40	0.00	0.00	0.00	0%	0%	
 Primary hand over		14	09-Oct-25	28-Oct-25	09-Oct-25	28-Oct-25	140,049.00	0.00	0.00	0.00	0%	0%	
 29	Primary hand over to the client	14	09-Oct-25	28-Oct-25	09-Oct-25	28-Oct-25	140,049.00	0.00	0.00	0.00	0%	0%	
					Page 1 of 1					TASK filter: All Activities			
										© Oracle Corporation			

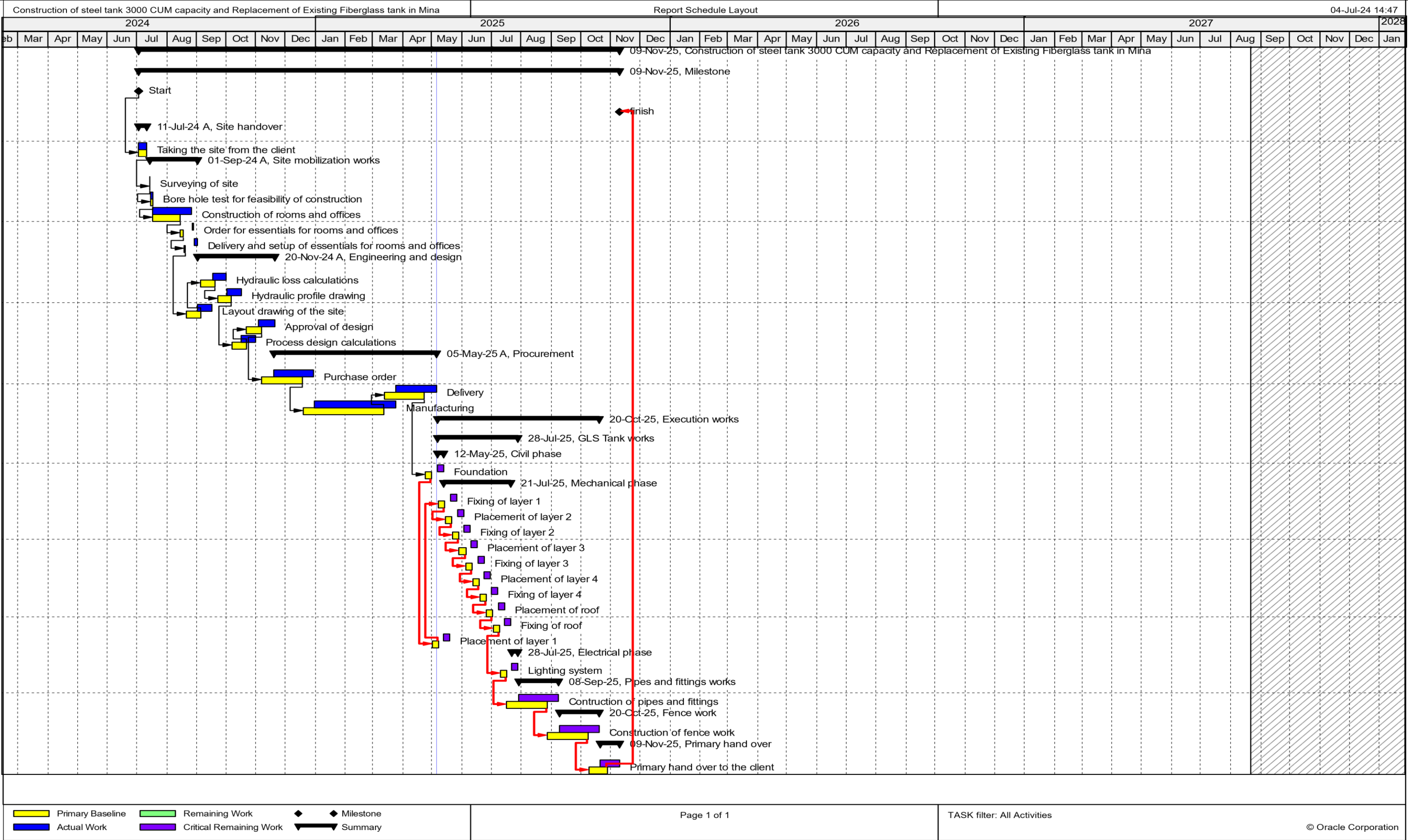


43 | Page

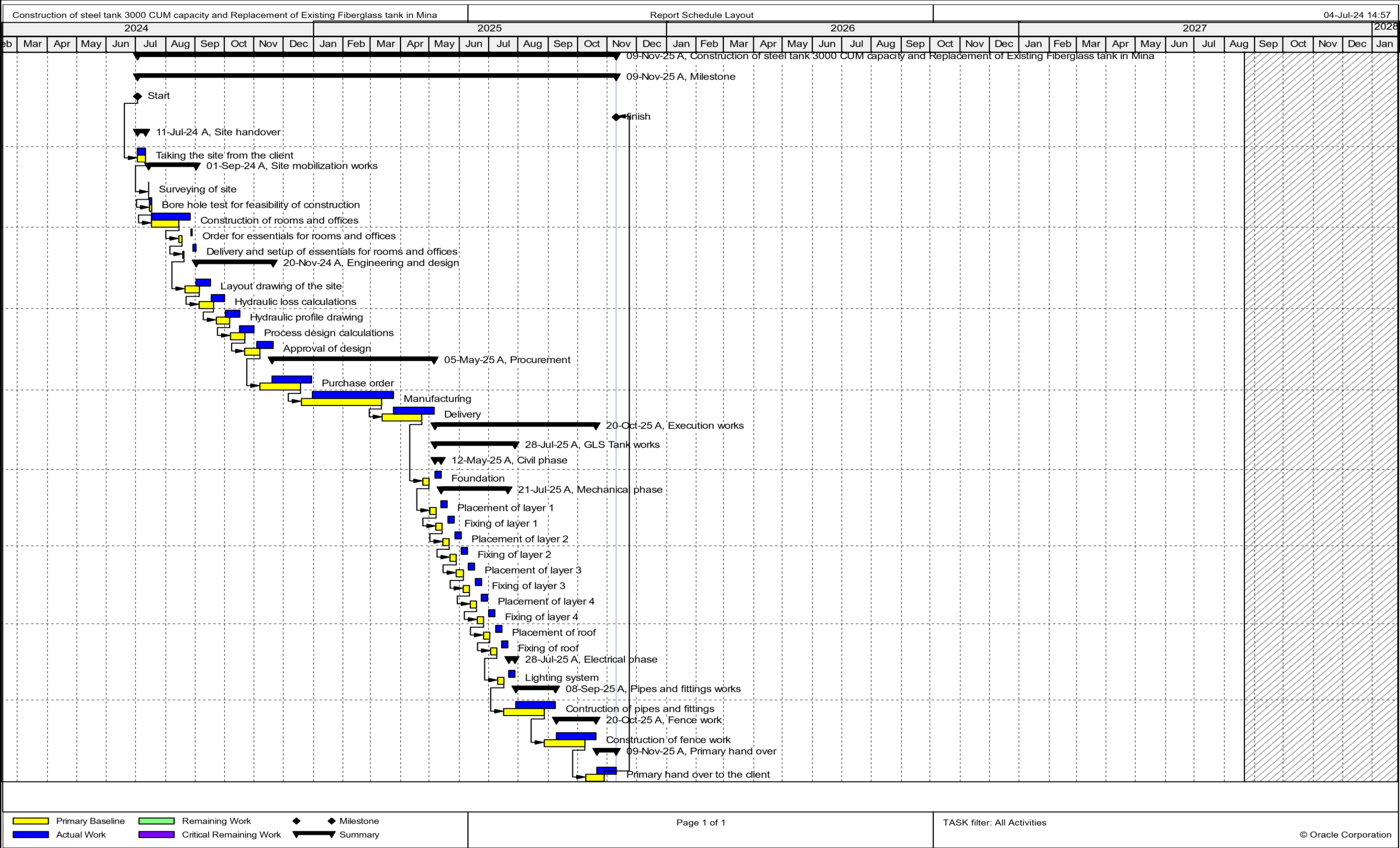
Construction of steel tank 3000 CUM capacity and Replacement of Existing Fiberglass tank in Mina				Report Schedule Layout					04-Jul-24 14:36			
Activity ID	Activity Name	Original Duration	Start	Finish	BL Project Start	BL Project Finish	Budgeted Total Cost	Actual Total Cost	Planned Value Cost	Earned Value Cost	Performance % Complete	Schedule % Complete
 Construction of steel tank 3000 CUM cap		353	03-Jul-24 A	09-Nov-25	03-Jul-24	28-Oct-25	52,977,441.70	0.00	0.00	0.00	0%	0%
 Milestone		353	03-Jul-24 A	09-Nov-25	03-Jul-24	28-Oct-25	0.00	0.00	0.00	0.00	0%	0%
 A	Start	0	03-Jul-24 A		03-Jul-24		0.00	0.00	0.00	0.00	100%	100%
 B	finish	0		09-Nov-25		28-Oct-25	0.00	0.00	0.00	0.00	0%	0%
 Site handover		7	03-Jul-24 A	11-Jul-24 ,	03-Jul-24	11-Jul-24	0.00	0.00	0.00	0.00	0%	0%
 1	Taking the site from the client	7	03-Jul-24 A	11-Jul-24 ,	03-Jul-24	11-Jul-24	0.00	0.00	0.00	0.00	100%	100%
 Site mobilization works		36	14-Jul-24 A	01-Sep-24	14-Jul-24	20-Aug-24	0.00	0.00	0.00	0.00	0%	0%
 2	Surveying of site	1	14-Jul-24 A	14-Jul-24 ,	14-Jul-24	14-Jul-24	0.00	0.00	0.00	0.00	100%	100%
 3	Bore hole test for feasibility of construction	3	15-Jul-24 A	17-Jul-24 ,	15-Jul-24	17-Jul-24	0.00	0.00	0.00	0.00	100%	100%
 4	Construction of rooms and offices	20	18-Jul-24 A	26-Aug-24	18-Jul-24	14-Aug-24	0.00	0.00	0.00	0.00	80%	100%
 5	Order for essentials for rooms and offices	2	27-Aug-24	28-Aug-24	15-Aug-24	18-Aug-24	0.00	0.00	0.00	0.00	0%	100%
 6	Delivery and setup of essentials for rooms and office:	2	29-Aug-24	01-Sep-24	19-Aug-24	20-Aug-24	0.00	0.00	0.00	0.00	0%	100%
 Engineering and design		56	02-Sep-24	18-Nov-24	21-Aug-24	06-Nov-24	21,770.80	0.00	0.00	0.00	0%	0%
 7	Layout drawing of the site	11	02-Sep-24	16-Sep-24	21-Aug-24	04-Sep-24	6,000.00	0.00	0.00	0.00	0%	0%
 8	Hydraulic loss calculations	11	17-Sep-24	01-Oct-24	05-Sep-24	19-Sep-24	1,200.00	0.00	0.00	0.00	0%	0%
 9	Hydraulic profile drawing	11	02-Oct-24	16-Oct-24	22-Sep-24	06-Oct-24	1,556.80	0.00	0.00	0.00	0%	0%
 10	Process design calculations	11	17-Oct-24	31-Oct-24	07-Oct-24	21-Oct-24	7,014.00	0.00	0.00	0.00	0%	0%
 11	Approval of design	12	03-Nov-24	18-Nov-24	22-Oct-24	06-Nov-24	6,000.00	0.00	0.00	0.00	0%	0%
 Procurement		120	19-Nov-24	05-May-25	07-Nov-24	23-Apr-25	12,547,345.00	0.00	0.00	0.00	0%	0%
 12	Purchase order	30	19-Nov-24	30-Dec-24	07-Nov-24	18-Dec-24	2,047,345.00	0.00	0.00	0.00	0%	0%
 13	Manufacturing	60	31-Dec-24	24-Mar-25	19-Dec-24	12-Mar-25	6,000,000.00	0.00	0.00	0.00	0%	0%
 14	Delivery	30	25-Mar-25	05-May-25	13-Mar-25	23-Apr-25	4,500,000.00	0.00	0.00	0.00	0%	0%
 Execution works		120	06-May-25	20-Oct-25	24-Apr-25	08-Oct-25	40,268,276.90	0.00	0.00	0.00	0%	0%
 GLS Tank works		60	06-May-25	28-Jul-25	24-Apr-25	16-Jul-25	38,385,727.00	0.00	0.00	0.00	0%	0%
 Civil phase		5	06-May-25	12-May-25	24-Apr-25	30-Apr-25	2,135,727.00	0.00	0.00	0.00	0%	0%
 15	Foundation	5	06-May-25	12-May-25	24-Apr-25	30-Apr-25	2,135,727.00	0.00	0.00	0.00	0%	0%
 Mechanical phase		50	13-May-25	21-Jul-25	01-May-25	09-Jul-25	33,750,000.00	0.00	0.00	0.00	0%	0%
 16	Placement of layer 1	5	13-May-25*	19-May-25	01-May-25	07-May-25	6,250,000.00	0.00	0.00	0.00	0%	0%
 17	Fixing of layer 1	5	20-May-25	26-May-25	08-May-25	14-May-25	500,000.00	0.00	0.00	0.00	0%	0%
 18	Placement of layer 2	5	27-May-25	02-Jun-25	15-May-25	21-May-25	6,250,000.00	0.00	0.00	0.00	0%	0%
 19	Fixing of layer 2	5	03-Jun-25	09-Jun-25	22-May-25	28-May-25	500,000.00	0.00	0.00	0.00	0%	0%
 20	Placement of layer 3	5	10-Jun-25	16-Jun-25	29-May-25	04-Jun-25	6,250,000.00	0.00	0.00	0.00	0%	0%
 21	Fixing of layer 3	5	17-Jun-25	23-Jun-25	05-Jun-25	11-Jun-25	500,000.00	0.00	0.00	0.00	0%	0%
 22	Placement of layer 4	5	24-Jun-25	30-Jun-25	12-Jun-25	18-Jun-25	6,250,000.00	0.00	0.00	0.00	0%	0%
 23	Fixing of layer 4	5	01-Jul-25	07-Jul-25	19-Jun-25	25-Jun-25	500,000.00	0.00	0.00	0.00	0%	0%
 24	Placement of roof	5	08-Jul-25	14-Jul-25	26-Jun-25	02-Jul-25	6,250,000.00	0.00	0.00	0.00	0%	0%
 25	Fixing of roof	5	15-Jul-25	21-Jul-25	03-Jul-25	09-Jul-25	500,000.00	0.00	0.00	0.00	0%	0%
 Electrical phase		5	22-Jul-25	28-Jul-25	10-Jul-25	16-Jul-25	2,500,000.00	0.00	0.00	0.00	0%	0%
 26	Lighting system	5	22-Jul-25	28-Jul-25	10-Jul-25	16-Jul-25	2,500,000.00	0.00	0.00	0.00	0%	0%
 Pipes and fittings works		30	29-Jul-25	08-Sep-25	17-Jul-25	27-Aug-25	1,528,499.50	0.00	0.00	0.00	0%	0%
 27	Contruction of pipes and fittings	30	29-Jul-25	08-Sep-25	17-Jul-25	27-Aug-25	1,528,499.50	0.00	0.00	0.00	0%	0%
 Fence work		30	09-Sep-25	20-Oct-25	28-Aug-25	08-Oct-25	354,050.40	0.00	0.00	0.00	0%	0%
 28	Construction of fence work	30	09-Sep-25	20-Oct-25	28-Aug-25	08-Oct-25	354,050.40	0.00	0.00	0.00	0%	0%
 Primary hand over		14	21-Oct-25	09-Nov-25	09-Oct-25	28-Oct-25	140,049.00	0.00	0.00	0.00	0%	0%
 29	Primary hand over to the client	14	21-Oct-25	09-Nov-25	09-Oct-25	28-Oct-25	140,049.00	0.00	0.00	0.00	0%	0%
				Page 1 of 1					TASK filter: All Activities			
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Construction of steel tank 3000 CUM capacity and Replacement of Existing Fiberglass tank in Mina				Report Schedule Layout					04-Jul-24 14:46			
Activity ID	Activity Name	Original Duration	Start	Finish	BL Project Start	BL Project Finish	Budgeted Total Cost	Actual Total Cost	Planned Value Cost	Earned Value Cost	Performance % Complete	Schedule % Complete
📁 Construction of steel tank 3000 CUM cap		353	03-Jul-24 A	09-Nov-25	03-Jul-24	28-Oct-25	52,977,441.70	12,563,115.80	18,454,842.80	12,569,115.80	23.73%	34.84%
📁 Milestone		353	03-Jul-24 A	09-Nov-25	03-Jul-24	28-Oct-25	0.00	0.00	0.00	0.00	0%	0%
📁 A	Start	0	03-Jul-24 A		03-Jul-24		0.00	0.00	0.00	0.00	100%	100%
📁 B	finish	0		09-Nov-25		28-Oct-25	0.00	0.00	0.00	0.00	0%	0%
📁 Site handover		7	03-Jul-24 A	11-Jul-24 ,	03-Jul-24	11-Jul-24	0.00	0.00	0.00	0.00	0%	0%
📁 1	Taking the site from the client	7	03-Jul-24 A	11-Jul-24 ,	03-Jul-24	11-Jul-24	0.00	0.00	0.00	0.00	100%	100%
📁 Site mobilization works		36	14-Jul-24 A	01-Sep-24	14-Jul-24	20-Aug-24	0.00	0.00	0.00	0.00	0%	0%
📁 2	Surveying of site	1	14-Jul-24 A	14-Jul-24 ,	14-Jul-24	14-Jul-24	0.00	0.00	0.00	0.00	100%	100%
📁 3	Bore hole test for feasibility of construction	3	15-Jul-24 A	17-Jul-24 ,	15-Jul-24	17-Jul-24	0.00	0.00	0.00	0.00	100%	100%
📁 4	Construction of rooms and offices	20	18-Jul-24 A	26-Aug-24	18-Jul-24	14-Aug-24	0.00	0.00	0.00	0.00	100%	100%
📁 5	Order for essentials for rooms and offices	2	27-Aug-24	28-Aug-24	15-Aug-24	18-Aug-24	0.00	0.00	0.00	0.00	100%	100%
📁 6	Delivery and setup of essentials for rooms and office:	2	29-Aug-24	01-Sep-24	19-Aug-24	20-Aug-24	0.00	0.00	0.00	0.00	100%	100%
📁 Engineering and design		56	02-Sep-24	20-Nov-24	21-Aug-24	06-Nov-24	21,770.80	15,770.80	21,770.80	21,770.80	100%	100%
📁 8	Hydraulic loss calculations	11	17-Sep-24	01-Oct-24	05-Sep-24	19-Sep-24	1,200.00	1,200.00	1,200.00	1,200.00	100%	100%
📁 9	Hydraulic profile drawing	11	02-Oct-24	16-Oct-24	22-Sep-24	06-Oct-24	1,556.80	1,556.80	1,556.80	1,556.80	100%	100%
📁 7	Layout drawing of the site	11	02-Sep-24	16-Sep-24	21-Aug-24	04-Sep-24	6,000.00	0.00	6,000.00	6,000.00	100%	100%
📁 11	Approval of design	12	03-Nov-24	20-Nov-24	22-Oct-24	06-Nov-24	6,000.00	6,000.00	6,000.00	6,000.00	100%	100%
📁 10	Process design calculations	11	17-Oct-24	31-Oct-24	07-Oct-24	21-Oct-24	7,014.00	7,014.00	7,014.00	7,014.00	100%	100%
📁 Procurement		120	19-Nov-24	05-May-25	07-Nov-24	23-Apr-25	12,547,345.00	12,547,345.00	12,547,345.00	12,547,345.00	100%	100%
📁 12	Purchase order	30	19-Nov-24	30-Dec-24	07-Nov-24	18-Dec-24	2,047,345.00	2,047,345.00	2,047,345.00	2,047,345.00	100%	100%
📁 14	Delivery	30	25-Mar-25	05-May-25	13-Mar-25	23-Apr-25	4,500,000.00	4,500,000.00	4,500,000.00	4,500,000.00	100%	100%
📁 13	Manufacturing	60	31-Dec-24	24-Mar-25	19-Dec-24	12-Mar-25	6,000,000.00	6,000,000.00	6,000,000.00	6,000,000.00	100%	100%
📁 Execution works		120	06-May-25	20-Oct-25	24-Apr-25	08-Oct-25	40,268,276.90	0.00	5,885,727.00	0.00	0%	14.62%
📁 GLS Tank works		60	06-May-25	28-Jul-25	24-Apr-25	16-Jul-25	38,385,727.00	0.00	5,885,727.00	0.00	0%	15.33%
📁 Civil phase		5	06-May-25	12-May-25	24-Apr-25	30-Apr-25	2,135,727.00	0.00	2,135,727.00	0.00	0%	100%
📁 15	Foundation	5	06-May-25	12-May-25	24-Apr-25	30-Apr-25	2,135,727.00	0.00	2,135,727.00	0.00	0%	100%
📁 Mechanical phase		50	13-May-25	21-Jul-25	01-May-25	09-Jul-25	33,750,000.00	0.00	3,750,000.00	0.00	0%	11.11%
📁 17	Fixing of layer 1	5	20-May-25	26-May-25	08-May-25	14-May-25	500,000.00	0.00	0.00	0.00	0%	0%
📁 18	Placement of layer 2	5	27-May-25	02-Jun-25	15-May-25	21-May-25	6,250,000.00	0.00	0.00	0.00	0%	0%
📁 19	Fixing of layer 2	5	03-Jun-25	09-Jun-25	22-May-25	28-May-25	500,000.00	0.00	0.00	0.00	0%	0%
📁 20	Placement of layer 3	5	10-Jun-25	16-Jun-25	29-May-25	04-Jun-25	6,250,000.00	0.00	0.00	0.00	0%	0%
📁 21	Fixing of layer 3	5	17-Jun-25	23-Jun-25	05-Jun-25	11-Jun-25	500,000.00	0.00	0.00	0.00	0%	0%
📁 22	Placement of layer 4	5	24-Jun-25	30-Jun-25	12-Jun-25	18-Jun-25	6,250,000.00	0.00	0.00	0.00	0%	0%
📁 23	Fixing of layer 4	5	01-Jul-25	07-Jul-25	19-Jun-25	25-Jun-25	500,000.00	0.00	0.00	0.00	0%	0%
📁 24	Placement of roof	5	08-Jul-25	14-Jul-25	26-Jun-25	02-Jul-25	6,250,000.00	0.00	0.00	0.00	0%	0%
📁 25	Fixing of roof	5	15-Jul-25	21-Jul-25	03-Jul-25	09-Jul-25	500,000.00	0.00	0.00	0.00	0%	0%
📁 16	Placement of layer 1	5	13-May-25*	19-May-25	01-May-25	07-May-25	6,250,000.00	0.00	3,750,000.00	0.00	0%	60%
📁 Electrical phase		5	22-Jul-25	28-Jul-25	10-Jul-25	16-Jul-25	2,500,000.00	0.00	0.00	0.00	0%	0%
📁 26	Lighting system	5	22-Jul-25	28-Jul-25	10-Jul-25	16-Jul-25	2,500,000.00	0.00	0.00	0.00	0%	0%
📁 Pipes and fittings works		30	29-Jul-25	08-Sep-25	17-Jul-25	27-Aug-25	1,528,499.50	0.00	0.00	0.00	0%	0%
📁 27	Contruction of pipes and fittings	30	29-Jul-25	08-Sep-25	17-Jul-25	27-Aug-25	1,528,499.50	0.00	0.00	0.00	0%	0%
📁 Fence work		30	09-Sep-25	20-Oct-25	28-Aug-25	08-Oct-25	354,050.40	0.00	0.00	0.00	0%	0%
📁 28	Construction of fence work	30	09-Sep-25	20-Oct-25	28-Aug-25	08-Oct-25	354,050.40	0.00	0.00	0.00	0%	0%
📁 Primary hand over		14	21-Oct-25	09-Nov-25	09-Oct-25	28-Oct-25	140,049.00	0.00	0.00	0.00	0%	0%
📁 29	Primary hand over to the client	14	21-Oct-25	09-Nov-25	09-Oct-25	28-Oct-25	140,049.00	0.00	0.00	0.00	0%	0%



Construction of steel tank 3000 CUM capacity and Replacement of Existing Fiberglass tank in Mina			Report Schedule Layout							04-Jul-24 15:07		
Activity ID	Activity Name	Original Duration	Start	Finish	BL Project Start	BL Project Finish	Budgeted Total Cost	Actual Total Cost	Planned Value Cost	Earned Value Cost	Performance % Complete	Schedule % Complete
 Construction of steel tank 3000 CUM ca		353	03-Jul-24 A	09-Nov-25 A	03-Jul-24	28-Oct-25	52,977,441.70	52,977,441.70	52,977,441.70	52,977,441.70	100%	100%
 Milestone		353	03-Jul-24 A	09-Nov-25 A	03-Jul-24	28-Oct-25	0.00	0.00	0.00	0.00	0%	0%
 A	Start	0	03-Jul-24 A		03-Jul-24		0.00	0.00	0.00	0.00	100%	100%
 B	finish	0		09-Nov-25 A		28-Oct-25	0.00	0.00	0.00	0.00	100%	100%
 Site handover		7	03-Jul-24 A	11-Jul-24 A	03-Jul-24	11-Jul-24	0.00	0.00	0.00	0.00	0%	0%
 1	Taking the site from the client	7	03-Jul-24 A	11-Jul-24 A	03-Jul-24	11-Jul-24	0.00	0.00	0.00	0.00	100%	100%
 Site mobilization works		36	14-Jul-24 A	01-Sep-24 A	14-Jul-24	20-Aug-24	0.00	0.00	0.00	0.00	0%	0%
 2	Surveying of site	1	14-Jul-24 A	14-Jul-24 A	14-Jul-24	14-Jul-24	0.00	0.00	0.00	0.00	100%	100%
 3	Bore hole test for feasibility of construction	3	15-Jul-24 A	17-Jul-24 A	15-Jul-24	17-Jul-24	0.00	0.00	0.00	0.00	100%	100%
 4	Construction of rooms and offices	20	18-Jul-24 A	26-Aug-24 A	18-Jul-24	14-Aug-24	0.00	0.00	0.00	0.00	100%	100%
 5	Order for essentials for rooms and offices	2	27-Aug-24 A	28-Aug-24 A	15-Aug-24	18-Aug-24	0.00	0.00	0.00	0.00	100%	100%
 6	Delivery and setup of essentials for rooms and office:	2	29-Aug-24 A	01-Sep-24 A	19-Aug-24	20-Aug-24	0.00	0.00	0.00	0.00	100%	100%
 Engineering and design		56	02-Sep-24 A	20-Nov-24 A	21-Aug-24	06-Nov-24	21,770.80	21,770.80	21,770.80	21,770.80	100%	100%
 7	Layout drawing of the site	11	02-Sep-24 A	16-Sep-24 A	21-Aug-24	04-Sep-24	6,000.00	6,000.00	6,000.00	6,000.00	100%	100%
 8	Hydraulic loss calculations	11	17-Sep-24 A	01-Oct-24 A	05-Sep-24	19-Sep-24	1,200.00	1,200.00	1,200.00	1,200.00	100%	100%
 9	Hydraulic profile drawing	11	02-Oct-24 A	16-Oct-24 A	22-Sep-24	06-Oct-24	1,556.80	1,556.80	1,556.80	1,556.80	100%	100%
 10	Process design calculations	11	17-Oct-24 A	31-Oct-24 A	07-Oct-24	21-Oct-24	7,014.00	7,014.00	7,014.00	7,014.00	100%	100%
 11	Approval of design	12	03-Nov-24 A	20-Nov-24 A	22-Oct-24	06-Nov-24	6,000.00	6,000.00	6,000.00	6,000.00	100%	100%
 Procurement		120	19-Nov-24 A	05-May-25 A	07-Nov-24	23-Apr-25	12,547,345.00	12,547,345.00	12,547,345.00	12,547,345.00	100%	100%
 12	Purchase order	30	19-Nov-24 A	30-Dec-24 A	07-Nov-24	18-Dec-24	2,047,345.00	2,047,345.00	2,047,345.00	2,047,345.00	100%	100%
 13	Manufacturing	60	31-Dec-24 A	24-Mar-25 A	19-Dec-24	12-Mar-25	6,000,000.00	6,000,000.00	6,000,000.00	6,000,000.00	100%	100%
 14	Delivery	30	25-Mar-25 A	05-May-25 A	13-Mar-25	23-Apr-25	4,500,000.00	4,500,000.00	4,500,000.00	4,500,000.00	100%	100%
 Execution works		120	06-May-25 A	20-Oct-25 A	24-Apr-25	08-Oct-25	40,268,276.90	40,268,276.90	40,268,276.90	40,268,276.90	100%	100%
 GLS Tank works		60	06-May-25 A	28-Jul-25 A	24-Apr-25	16-Jul-25	38,385,727.00	38,385,727.00	38,385,727.00	38,385,727.00	100%	100%
 Civil phase		5	06-May-25 A	12-May-25 A	24-Apr-25	30-Apr-25	2,135,727.00	2,135,727.00	2,135,727.00	2,135,727.00	100%	100%
 15	Foundation	5	06-May-25 A	12-May-25 A	24-Apr-25	30-Apr-25	2,135,727.00	2,135,727.00	2,135,727.00	2,135,727.00	100%	100%
 Mechanical phase		50	13-May-25 A	21-Jul-25 A	01-May-25	09-Jul-25	33,750,000.00	33,750,000.00	33,750,000.00	33,750,000.00	100%	100%
 16	Placement of layer 1	5	13-May-25 A	19-May-25 A	01-May-25	07-May-25	6,250,000.00	6,250,000.00	6,250,000.00	6,250,000.00	100%	100%
 17	Fixing of layer 1	5	20-May-25 A	26-May-25 A	08-May-25	14-May-25	500,000.00	500,000.00	500,000.00	500,000.00	100%	100%
 18	Placement of layer 2	5	27-May-25 A	02-Jun-25 A	15-May-25	21-May-25	6,250,000.00	6,250,000.00	6,250,000.00	6,250,000.00	100%	100%
 19	Fixing of layer 2	5	03-Jun-25 A	09-Jun-25 A	22-May-25	28-May-25	500,000.00	500,000.00	500,000.00	500,000.00	100%	100%
 20	Placement of layer 3	5	10-Jun-25 A	16-Jun-25 A	29-May-25	04-Jun-25	6,250,000.00	6,250,000.00	6,250,000.00	6,250,000.00	100%	100%
 21	Fixing of layer 3	5	17-Jun-25 A	23-Jun-25 A	05-Jun-25	11-Jun-25	500,000.00	500,000.00	500,000.00	500,000.00	100%	100%
 22	Placement of layer 4	5	24-Jun-25 A	30-Jun-25 A	12-Jun-25	18-Jun-25	6,250,000.00	6,250,000.00	6,250,000.00	6,250,000.00	100%	100%
 23	Fixing of layer 4	5	01-Jul-25 A	07-Jul-25 A	19-Jun-25	25-Jun-25	500,000.00	500,000.00	500,000.00	500,000.00	100%	100%
 24	Placement of roof	5	08-Jul-25 A	14-Jul-25 A	26-Jun-25	02-Jul-25	6,250,000.00	6,250,000.00	6,250,000.00	6,250,000.00	100%	100%
 25	Fixing of roof	5	15-Jul-25 A	21-Jul-25 A	03-Jul-25	09-Jul-25	500,000.00	500,000.00	500,000.00	500,000.00	100%	100%
 Electrical phase		5	22-Jul-25 A	28-Jul-25 A	10-Jul-25	16-Jul-25	2,500,000.00	2,500,000.00	2,500,000.00	2,500,000.00	100%	100%
 26	Lighting system	5	22-Jul-25 A	28-Jul-25 A	10-Jul-25	16-Jul-25	2,500,000.00	2,500,000.00	2,500,000.00	2,500,000.00	100%	100%
 Pipes and fittings works		30	29-Jul-25 A	08-Sep-25 A	17-Jul-25	27-Aug-25	1,528,499.50	1,528,499.50	1,528,499.50	1,528,499.50	100%	100%
 27	Contruction of pipes and fittings	30	29-Jul-25 A	08-Sep-25 A	17-Jul-25	27-Aug-25	1,528,499.50	1,528,499.50	1,528,499.50	1,528,499.50	100%	100%
 Fence work		30	09-Sep-25 A	20-Oct-25 A	28-Aug-25	08-Oct-25	354,050.40	354,050.40	354,050.40	354,050.40	100%	100%
 28	Construction of fence work	30	09-Sep-25 A	20-Oct-25 A	28-Aug-25	08-Oct-25	354,050.40	354,050.40	354,050.40	354,050.40	100%	100%
 Primary hand over		14	21-Oct-25 A	09-Nov-25 A	09-Oct-25	28-Oct-25	140,049.00	140,049.00	140,049.00	140,049.00	100%	100%
 29	Primary hand over to the client	14	21-Oct-25 A	09-Nov-25 A	09-Oct-25	28-Oct-25	140,049.00	140,049.00	140,049.00	140,049.00	100%	100%



6.3 Resource Allocation

- in Primavera P6, resource allocation refers to the process of assigning available resources (such as labour, equipment, or materials) to project activities to ensure efficient project execution. Effective resource allocation is critical for meeting project timelines, managing costs, and preventing overallocation.
- Here's a breakdown of how it works in Primavera P6:
 - **Create Resource Pool:**

In Primavera, resources are first defined in a resource pool with details like hourly rates, resource type (labour, material, or non-labour), and availability.
 - **Assign Resource Calendars:**

Resources are assigned specific calendars that define their working hours and availability (e.g., full-time, part-time, or shift-based), which Primavera uses to calculate their work periods on activities.
 - **Resource Assignment:**

Resources are assigned to individual activities in the project's Work Breakdown Structure (WBS). Primavera considers each resource's availability when calculating activity durations and costs.
 - **Resource Loading:**

Primavera allows you to load resources based on units (hours or quantities). The load rate (e.g., hours per day) defines how intensively a resource is applied to an activity.
- Given ahead is an example of a task given about a project on "Al Batha", where I was assigned a task of assigning resource on primavera using values given.

			11-Aug-24 02:21			
Activity ID	Resource ID	Activity Name	Start	Finish	Price / Unit	Budgeted Units
Batha			21-Jul-24	09-Jul-25		14732
supplying, and construct Top course			08-Jul-25	09-Jul-25		350
109	AB-109	supplying, and construct , asphaltting, , paving and testing of asphalt yards and roads with a layer of asphalt (Top)	08-Jul-25	09-Jul-25	75.00	350
supplying, and construct Binder course			06-Jul-25	07-Jul-25		350
108	AB-108	supplying, and construct , asphaltting, , paving and testing of asphalt yards and roads with a layer of asphalt (Binder)	06-Jul-25	07-Jul-25	75.00	350
improved gravel soil			02-Jul-25	03-Jul-25		350
107	AB-107	Supplying and testing a layer of improved gravel soil with a granular gradation of 200 mm thickness, class A-1-A	02-Jul-25	03-Jul-25	70.00	350
Site preparation works			30-Jun-25	01-Jul-25		350
106	AB-106	Site preparation works - Preparing and leveling and Grading of subgrade	30-Jun-25	01-Jul-25	50.00	350
Rehabilitation of the Generator (New battery + maintainance for MTS , change oil , change all filters			26-Jun-25	29-Jun-25		1
105	AB-105	Rehabilitation of the Generator (New battery + maintainance for MTS , change oil , change all filters	26-Jun-25	29-Jun-25	90000.00	1
Rehabilitation of Cable tray (Existing)			24-Jun-25	25-Jun-25		20
104	AB-104	Rehabilitation of Cable tray (Existing)	24-Jun-25	25-Jun-25	756.00	20
Supply, install, test and delivery of Pipes and manholes work for electrical works the work included new UPCV			22-Jun-25	23-Jun-25		360
103	AB-103	Supply, install, test and delivery of Pipes and manholes work for electrical works the work included new UPCV	22-Jun-25	23-Jun-25	258.00	360
Supply, installation, testing and delivery of Main cables 3*240mm +120G+120E CU/XLPE/PVC			18-Jun-25	19-Jun-25		60
102	AB-102	Supply, installation, testing and delivery of Main cables 3*240mm +120G+120E CU/XLPE/PVC	18-Jun-25	19-Jun-25	714.00	60
Supply, installation, testing and delivery of Cables between units CU/PVC/PVC inside MCC room			16-Jun-25	17-Jun-25		300
101	AB-101	Supply, installation, testing and delivery of Cables between units CU/PVC/PVC inside MCC room	16-Jun-25	17-Jun-25	450.00	300
Rehabilitation of Electrical control panel			12-Jun-25	15-Jun-25		1
100	AB-100	Rehabilitation of Electrical control panel	12-Jun-25	15-Jun-25	159000.00	1
Supply, installation, testing and delivery of piping and fittings UPVC Pipes (200-50mm), PN 10 install on TSE			10-Jun-25	11-Jun-25		1
99	AB-99	Supply, installation, testing and delivery of piping and fittings UPVC Pipes (200-50mm), PN 10 install on TSE	10-Jun-25	11-Jun-25	86000.00	1
Execution Epoxy 2 layers for TSE (product / treated) tank GLS foundation (Diameter: 17.85 m Height: 4.23 m)			08-Jun-25	09-Jun-25		1
98	AB-98	Execution Epoxy 2 layers for TSE (product / treated) tank GLS foundation (Diameter: 17.85 m Height: 4.23 m)	08-Jun-25	09-Jun-25	52000.00	1
Rehabilitation of the TSE (product / treated) Tank GLS Existing nozzles, plate form, ladder by Epoxy paint, bolts			04-Jun-25	05-Jun-25		1
97	AB-97	Rehabilitation of the TSE (product / treated) Tank GLS Existing nozzles, plate form, ladder by Epoxy paint, bolts	04-Jun-25	05-Jun-25	35000.00	1
Rehabilitation Existing TSE (product / treated) Tank GLS Diameter : 17.85 m			02-Jun-25	03-Jun-25		1
96	AB-96	Rehabilitation Existing TSE (product / treated) Tank GLS Diameter : 17.85 m	02-Jun-25	03-Jun-25	145500.00	1
Supply, installation, testing and delivery of piping and fittings UPVC Pipes (200-50mm), PN 10			29-May-25	01-Jun-25		1
95	AB-95	Supply, installation, testing and delivery of piping and fittings UPVC Pipes (200-50mm), PN 10	29-May-25	01-Jun-25	59000.00	1
Execution Epoxy 2 layers for Sludge Holding tank GLS foundation (Diameter : 6.80 m Height : 4.23 m)			27-May-25	28-May-25		1
94	AB-94	Execution Epoxy 2 layers for Sludge Holding tank GLS foundation (Diameter : 6.80 m Height : 4.23 m)	27-May-25	28-May-25	29000.00	1
Rehabilitation of the Sludge Holding tank GLS Existing nozzles, plate form , ladder by Epoxy paint, bolts			25-May-25	26-May-25		1
93	AB-93	Rehabilitation of the Sludge Holding tank GLS Existing nozzles, plate form , ladder by Epoxy paint, bolts	25-May-25	26-May-25	28000.00	1
Rehabilitation Existing Sludge Holding tank GLS Diameter: 6.80 m			21-May-25	22-May-25		1
92	AB-92	Rehabilitation Existing Sludge Holding tank GLS Diameter: 6.80 m	21-May-25	22-May-25	49000.00	1
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							11-Aug-24 02:21	
Activity ID	Resource ID	Activity Name	Start	Finish	Price / Unit	Budgeted Units		
		Supply, installation and testing of piping and fittings UPVC Pipes (200-50mm), PN 10 install on Chlorine Contact tank	19-May-25	20-May-25		1		
91	AB-91	Supply, installation and testing of piping and fittings UPVC Pipes (200-50mm), PN 10 install on Chlorine Contact tank	19-May-25	20-May-25	58000.00	1		
		Execution Epoxy 2 layers for Chlorine Contact tank GLS foundation (Diameter : 11.05 m Height : 2.83 m)	15-May-25	18-May-25		1		
90	AB-90	Execution Epoxy 2 layers for Chlorine Contact tank GLS foundation (Diameter : 11.05 m Height : 2.83 m)	15-May-25	18-May-25	36000.00	1		
		Rehabilitation of the Chlorine Contact tank GLS Existing nozzles, plate form, ladder by Epoxy paint, bolts	13-May-25	14-May-25		1		
89	AB-89	Rehabilitation of the Chlorine Contact tank GLS Existing nozzles, plate form, ladder by Epoxy paint, bolts	13-May-25	14-May-25	12000.00	1		
		Rehabilitation of Existing Chlorine Contact tank GLS Diameter: 11.05 m Height : 2.83 m	11-May-25	12-May-25		1		
88	AB-88	Rehabilitation of Existing Chlorine Contact tank GLS Diameter: 11.05 m Height : 2.83 m	11-May-25	12-May-25	61000.00	1		
		Remove old (Existing) GLS Settlement tank	07-May-25	08-May-25		1		
87	AB-87	Remove old (Existing) GLS Settlement tank	07-May-25	08-May-25	88000.00	1		
		Execution RCC Tank (tank foundation) 11.94 m dia the work included Epoxy works	05-May-25	06-May-25		1		
86	AB-86	Execution RCC Tank (tank foundation) 11.94 m dia the work included Epoxy works	05-May-25	06-May-25	164000.00	1		
		Supply, installation, testing and delivery of piping and fittings UPVC Pipes (200-50mm) PN 10 on Settlement tank	01-May-25	04-May-25		1		
85	AB-85	Supply, installation, testing and delivery of piping and fittings UPVC Pipes (200-50mm) PN 10 on Settlement tank	01-May-25	04-May-25	53000.00	1		
		Supply, install, test and delivery of New GLS Tank 11.94 m dia. x 4.26 m H with all accessories	29-Apr-25	30-Apr-25		1		
84	AB-84	Supply, install, test and delivery of New GLS Tank 11.94 m dia. x 4.26 m H with all accessories	29-Apr-25	30-Apr-25	368000.00	1		
		Execution Epoxy 2 layers for areation Tank foundation (Diameter : 18.70 m)	27-Apr-25	28-Apr-25		1		
83	AB-83	Execution Epoxy 2 layers for areation Tank foundation (Diameter : 18.70 m)	27-Apr-25	28-Apr-25	46000.00	1		
		Rehabilitation of the Existing nozzles, plate form, ladder by Epoxy paint, bolts	23-Apr-25	24-Apr-25		1		
82	AB-82	Rehabilitation of the Existing nozzles, plate form, ladder by Epoxy paint, bolts	23-Apr-25	24-Apr-25	30000.00	1		
		Rehabilitation Existing aration tank Diameter: 18.70 m, Height : 4.23 m	21-Apr-25	22-Apr-25		1		
81	AB-81	Rehabilitation Existing aration tank Diameter: 18.70 m, Height : 4.23 m	21-Apr-25	22-Apr-25	54000.00	1		
		Supply, installation, testing and delivery of CLARIFIER	17-Apr-25	20-Apr-25		1		
80	AB-80	Supply, installation, testing and delivery of CLARIFIER	17-Apr-25	20-Apr-25	211000.00	1		
		Supply, installation, testing and delivery of Type: Belt Filter Press	15-Apr-25	16-Apr-25		1		
79	AB-79	Supply, installation, testing and delivery of Type: Belt Filter Press	15-Apr-25	16-Apr-25	703000.00	1		
		Supply, install, test and delivery of CS Filter Diamter 3.0 m Total Height : 2.50 m	13-Apr-25	14-Apr-25		1		
78	AB-78	Supply, install, test and delivery of CS Filter Diamter 3.0 m Total Height : 2.50 m	13-Apr-25	14-Apr-25	182450.00	1		
		Supply, installation, testing and delivery of Dosing Pump 7.0 l/h @ 3.0 bar with all accessory	09-Apr-25	10-Apr-25		1		
77	AB-77	Supply, installation, testing and delivery of Dosing Pump 7.0 l/h @ 3.0 bar with all accessory	09-Apr-25	10-Apr-25	1580.00	1		
		Supply, install, test and delivery of Dosing Pump 7.0 l/h @ 3.0 bar with all accessory	07-Apr-25	08-Apr-25		1		
76	AB-76	Supply, install, test and delivery of Dosing Pump 7.0 l/h @ 3.0 bar with all accessory	07-Apr-25	08-Apr-25	1832.00	1		
		Supply, installation and testing of piping and fittings inside Equipment Room UPVC Pipes (300-50mm), PN, 10 , 1	03-Apr-25	06-Apr-25		1		
75	AB-75	Supply, installation and testing of piping and fittings inside Equipment Room UPVC Pipes (300-50mm), PN, 10 , 16	03-Apr-25	06-Apr-25	3000.00	1		
		Supply, install, test and delivery of Filter Feed & Backwash pump 167 m3/h @ 40m head(1 duty - 1 - standby)	01-Apr-25	02-Apr-25		2		
74	AB-74	Supply, install, test and delivery of Filter Feed & Backwash pump 167 m3/h @ 40m head(1 duty - 1 - standby)	01-Apr-25	02-Apr-25	23600.00	2		
		Supply, install, test and delivery of Air lift system the work included all material (GI -pvc) pipes, fittings, valves	30-Mar-25	31-Mar-25		1		
© Oracle Corporation			Page 2 of 6					

11-Aug-24 02:21						
Activity ID	Resource ID	Activity Name	Start	Finish	Price / Unit	Budgeted Units
73	AB-73	Supply, install, test and delivery of Air lift system the work included all material (GI -pvc) pipes, fittings, valves	30-Mar-25	31-Mar-25	40000.00	1
		Supply, install, test and delivery of air diffuser Fine Bubble Disc Type Membrane Diffusers, EPDM, PVC, Membrane	26-Mar-25	27-Mar-25		1
72	AB-72	Supply, install, test and delivery of air diffuser Fine Bubble Disc Type Membrane Diffusers, EPDM, PVC, Membrane Size	26-Mar-25	27-Mar-25	81500.00	1
		Supply, install, test and delivery of Air pressure Gage rang (0-4 bar) on pipe	24-Mar-25	25-Mar-25		2
71	AB-71	Supply, install, test and delivery of Air pressure Gage rang (0-4 bar) on pipe	24-Mar-25	25-Mar-25	500.00	2
		Supply, install, testing and delivery of DI GGG40 Check valve for air DN 150 PN 10	20-Mar-25	23-Mar-25		2
70	AB-70	Supply, install, testing and delivery of DI GGG40 Check valve for air DN 150 PN 10	20-Mar-25	23-Mar-25	2000.00	2
		Supply, install, test and delivery of DI GGG40 gate valve for air non rising steam DN 150, PN10	18-Mar-25	19-Mar-25		2
69	AB-69	Supply, install, test and delivery of DI GGG40 gate valve for air non rising steam DN 150, PN10	18-Mar-25	19-Mar-25	1500.00	2
		Supply, install, test and delivery of GI Piping and fitting DN (150) 6 inches- PN 10 - class 5	16-Mar-25	17-Mar-25		1
68	AB-68	Supply, install, test and delivery of GI Piping and fitting DN (150) 6 inches- PN 10 - class 5	16-Mar-25	17-Mar-25	15000.00	1
		Supply, install, test and delivery of Air blower Indoors type 2000 Nm3/h @ 0.5 bar (1 duty - standby)	12-Mar-25	13-Mar-25		2
67	AB-67	Supply, install, test and delivery of Air blower Indoors type 2000 Nm3/h @ 0.5 bar (1 duty - standby)	12-Mar-25	13-Mar-25	185000.00	2
		Supply, installation, testing and delivery of float Level Switch 220 V	10-Mar-25	11-Mar-25		3
66	AB-66	Supply, installation, testing and delivery of float Level Switch 220 V	10-Mar-25	11-Mar-25	200.00	3
		Supply, installation, testing and delivery of DI GGG40 Check valve DN 150 in valve chamber	06-Mar-25	09-Mar-25		3
65	AB-65	Supply, installation, testing and delivery of DI GGG40 Check valve DN 150 in valve chamber	06-Mar-25	09-Mar-25	2000.00	3
		Supply, install, test and commission gate valves (DI GGG40 gate valve , PN 16) - DN150 in valve chamber	04-Mar-25	05-Mar-25		5
64	AB-64	Supply, install, test and commission gate valves (DI GGG40 gate valve , PN 16) - DN150 in valve chamber	04-Mar-25	05-Mar-25	1700.00	5
		install, supply, test, delivery of submersible pump Q = 166.67 m3/h @Head 20 m(1 duty - 1 - standby)	02-Mar-25	03-Mar-25		3
63	AB-63	install, supply, test, delivery of submersible pump Q = 166.67 m3/h @Head 20 m(1 duty - 1 - standby)	02-Mar-25	03-Mar-25	61000.00	3
		Supply, installation, testing and delivery of Electromagnetic flowmeter DN 150	26-Feb-25	27-Feb-25		1
62	AB-62	Supply, installation, testing and delivery of Electromagnetic flowmeter DN 150	26-Feb-25	27-Feb-25	22000.00	1
		Rehabilitation BELT PRESS SYSTEM Room ,as per Existing room	24-Feb-25	25-Feb-25		1
61	AB-61	Rehabilitation BELT PRESS SYSTEM Room ,as per Existing room	24-Feb-25	25-Feb-25	7700.00	1
		Rehabilitation plant gate the work included fix ,paint	20-Feb-25	23-Feb-25		1
60	AB-60	Rehabilitation plant gate the work included fix ,paint	20-Feb-25	23-Feb-25	3000.00	1
		Rehabilitation of paint for Equipment Building	18-Feb-25	19-Feb-25		100
59	AB-59	Rehabilitation of paint for Equipment Building	18-Feb-25	19-Feb-25	30.00	100
		Supply, install, test and delivery of low voltage cable 4x50mm+16mm CU/XLPE/PVC	13-Feb-25	17-Feb-25		50
58	AB-58	Supply, install, test and delivery of low voltage cable 4x50mm+16mm CU/XLPE/PVC	13-Feb-25	17-Feb-25	300.00	50
		Supply, install, test and delivery of Control panel including contactors, relays ,switches and indicator lamps	10-Feb-25	12-Feb-25		1
57	AB-57	Supply, install, test and delivery of Control panel including contactors, relays ,switches and indicator lamps	10-Feb-25	12-Feb-25	40103.00	1
		Supply, installation, testing and delivery in a good working condition float Level Switch 220 V	05-Feb-25	09-Feb-25		3
56	AB-56	Supply, installation, testing and delivery in a good working condition float Level Switch 220 V	05-Feb-25	09-Feb-25	200.00	3
		Supply and installation of square ductile cover with dimensions (1000 mm x 1000 mm)	02-Feb-25	04-Feb-25		2
55	AB-55	Supply and installation of square ductile cover with dimensions (1000 mm x 1000 mm)	02-Feb-25	04-Feb-25	2150.00	2
© Oracle Corporation			Page 3 of 6			

11-Aug-24 02:21

Activity ID	Resource ID	Activity Name	Start	Finish	Price / Unit	Budgeted Units
 Interior floors Epoxy			28-Jan-25	30-Jan-25		24
 54 AB-54	AB-54	Interior floors Epoxy	28-Jan-25	30-Jan-25	60.00	24
 Interior walls Epoxy			23-Jan-25	27-Jan-25		130
 53 AB-53	AB-53	Interior walls Epoxy	23-Jan-25	27-Jan-25	60.00	130
 water stop width 200 mm as per drawings approved			20-Jan-25	22-Jan-25		20
 52 AB-52	AB-52	water stop width 200 mm as per drawings approved	20-Jan-25	22-Jan-25	50.00	20
 Surface preparation, application of bituminous as per drawings approved			15-Jan-25	19-Jan-25		130
 51 AB-51	AB-51	Surface preparation, application of bituminous as per drawings approved	15-Jan-25	19-Jan-25	30.00	130
 Polyethylene vapor barrier (250) below plain concrete			12-Jan-25	14-Jan-25		25
 50 AB-50	AB-50	Polyethylene vapor barrier (250) below plain concrete	12-Jan-25	14-Jan-25	30.00	25
 Reinforced concrete solid slab			07-Jan-25	09-Jan-25		7
 49 AB-49	AB-49	Reinforced concrete solid slab	07-Jan-25	09-Jan-25	2000.00	7
 Reinforced concrete for walls			02-Jan-25	06-Jan-25		36
 48 AB-48	AB-48	Reinforced concrete for walls	02-Jan-25	06-Jan-25	2000.00	36
 Reinforced concrete for foundation			30-Dec-24	01-Jan-25		7
 47 AB-47	AB-47	Reinforced concrete for foundation	30-Dec-24	01-Jan-25	2000.00	7
 Cast plain concrete THICKNESS 100 mm			25-Dec-24	29-Dec-24		3
 46 AB-46	AB-46	Cast plain concrete THICKNESS 100 mm	25-Dec-24	29-Dec-24	800.00	3
 Backfilling using Excavated Material			22-Dec-24	24-Dec-24		62
 45 AB-45	AB-45	Backfilling using Excavated Material	22-Dec-24	24-Dec-24	45.00	62
 Excavation from natural ground level			17-Dec-24	19-Dec-24		160
 44 AB-44	AB-44	Excavation from natural ground level	17-Dec-24	19-Dec-24	45.00	160
 drian Valve chamber length 1.3 m, W = 1.10, H = 1.90			12-Dec-24	16-Dec-24		2
 43 AB-43	AB-43	drian Valve chamber length 1.3 m, W = 1.10, H = 1.90	12-Dec-24	16-Dec-24	11000.00	2
 airless Valve chamber length 1.30 m , W = 1.35 , H = 1.7			09-Dec-24	11-Dec-24		2
 42 AB-42	AB-42	airless Valve chamber length 1.30 m , W = 1.35 , H = 1.7	09-Dec-24	11-Dec-24	10000.00	2
 manhole Depth up 3m			04-Dec-24	08-Dec-24		5
 41 AB-41	AB-41	manhole Depth up 3m	04-Dec-24	08-Dec-24	5800.00	5
 manhole Depth 2 m - 3m			01-Dec-24	03-Dec-24		16
 40 AB-40	AB-40	manhole Depth 2 m - 3m	01-Dec-24	03-Dec-24	5500.00	16
 manhole Depth 1.2 m- 2 m			26-Nov-24	28-Nov-24		9
 39 AB-39	AB-39	manhole Depth 1.2 m- 2 m	26-Nov-24	28-Nov-24	5000.00	9
 manhole Depth 1.2 m			21-Nov-24	25-Nov-24		20
 38 AB-38	AB-38	manhole Depth 1.2 m	21-Nov-24	25-Nov-24	4400.00	20
 submersible pump 45 m3/h @Head 65 m			18-Nov-24	20-Nov-24		2
 37 AB-37	AB-37	submersible pump 45 m3/h @Head 65 m	18-Nov-24	20-Nov-24	90000.00	2
 air valve DN 80			13-Nov-24	17-Nov-24		2

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Page 4 of 6

11-Aug-24 02:21							
Activity ID	Resource ID	Activity Name	Start	Finish	Price / Unit	Budgeted Units	
 36	AB-36	air valve DN 80	13-Nov-24	17-Nov-24	4000.00	2	
 check valve DN 150			10-Nov-24	12-Nov-24		2	
 35	AB-35	check valve DN 150	10-Nov-24	12-Nov-24	2000.00	2	
 gate valve DN 150			05-Nov-24	07-Nov-24		7	
 34	AB-34	gate valve DN 150	05-Nov-24	07-Nov-24	1500.00	7	
 pipe DN150 mm			31-Oct-24	04-Nov-24		4360	
 33	AB-33	pipe DN150 mm	31-Oct-24	04-Nov-24	89.00	4360	
 pipe DN200			28-Oct-24	30-Oct-24		3275	
 32	AB-32	pipe DN200	28-Oct-24	30-Oct-24	77.00	3275	
 low voltage cable 4x50mm+16mm CU/XLPE/PVC , included upvc pipes class 3			23-Oct-24	27-Oct-24		50	
 31	AB-31	low voltage cable 4x50mm+16mm CU/XLPE/PVC , included upvc pipes class 3	23-Oct-24	27-Oct-24	250.00	50	
 Control panel including contactors, relays ,switches and indicator lamps			20-Oct-24	22-Oct-24		1	
 30	AB-30	Control panel including contactors, relays ,switches and indicator lamps	20-Oct-24	22-Oct-24	36102.00	1	
 Float Level Switch 220 V			15-Oct-24	17-Oct-24		3	
 29	AB-29	Float Level Switch 220 V	15-Oct-24	17-Oct-24	270.00	3	
 Drain Valve chamber length 1.3 m, W = 1.10, H = 1.90			10-Oct-24	14-Oct-24		2	
 28	AB-28	Drian Valve chamber length 1.3 m, W = 1.10, H = 1.90	10-Oct-24	14-Oct-24	11000.00	2	
 Airless Valve chamber length 1.30 m , W = 1.35 , H = 1.7			07-Oct-24	09-Oct-24		1	
 27	AB-27	Airless Valve chamber length 1.30 m , W = 1.35 , H = 1.7	07-Oct-24	09-Oct-24	10000.00	1	
 Supply and installation of a square ductile cover with dimensions (1000mm*1000mm)			02-Oct-24	06-Oct-24		2	
 26	AB-26	Supply and installation of a square ductile cover with dimensions (1000mm*1000mm)	02-Oct-24	06-Oct-24	1735.00	2	
 Intrrior floors Epoxy			29-Sep-24	01-Oct-24		16	
 25	AB-25	Intrrior floors Epoxy	29-Sep-24	01-Oct-24	60.00	16	
 Interior walls Epoxy			24-Sep-24	26-Sep-24		128	
 24	AB-24	Interior walls Epoxy	24-Sep-24	26-Sep-24	60.00	128	
 water stop width 200 mm as per drawings approved			19-Sep-24	23-Sep-24		16	
 23	AB-23	water stop width 200 mm as per drawings approved	19-Sep-24	23-Sep-24	50.00	16	
 Surface preparation, application of bituminous as per drawings approved			16-Sep-24	18-Sep-24		133	
 22	AB-22	Surface preparation, application of bituminous as per drawings approved	16-Sep-24	18-Sep-24	30.00	133	
 Polyethylene vapor barrier (250) below plain concrete			11-Sep-24	15-Sep-24		20	
 21	AB-21	Polyethylene vapor bamer (250) below plain concrete	11-Sep-24	15-Sep-24	30.00	20	
 Reinforced concrete solid slab			08-Sep-24	10-Sep-24		5	
 20	AB-20	Reinforced concrete solid slab	08-Sep-24	10-Sep-24	2100.00	5	
 Reinforced concrete for walls			03-Sep-24	05-Sep-24		38	
 19	AB-19	Reinforced concrete for walls	03-Sep-24	05-Sep-24	2100.00	38	
 Reinforced concrete for foundation			29-Aug-24	02-Sep-24		5	
 18	AB-18	Reinforced concrete for foundation	29-Aug-24	02-Sep-24	2100.00	5	
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11-Aug-24 02:21

Activity ID	Resource ID	Activity Name	Start	Finish	Price / Unit	Budgeted Units
 Contrete of thickness 100 mm			26-Aug-24	28-Aug-24		2
 17	AB-17	Contrete of thickness 100 mm	26-Aug-24	28-Aug-24	800.00	2
 Backfilling using Excavated Material			21-Aug-24	25-Aug-24		50
 16	AB-16	Backfilling using Excavated Material	21-Aug-24	25-Aug-24	60.00	50
 Excavation from natural ground level			18-Aug-24	20-Aug-24		130
 15	AB-15	Excavation from natural ground level	18-Aug-24	20-Aug-24	60.00	130
 Depth greater than 3.0 and up to 4.0 m			14-Aug-24	15-Aug-24		7
 14	AB-14	Depth greater than 3.0 and up to 4.0 m	14-Aug-24	15-Aug-24	6000.00	7
 Depth greater than 2.0 and up to 3.0 m			12-Aug-24	13-Aug-24		5
 13	AB-13	Depth greater than 2.0 and up to 3.0 m	12-Aug-24	13-Aug-24	4200.00	5
 Depth greater than 1.2and up to 2.0 m			08-Aug-24	11-Aug-24		4
 12	AB-12	Depth greater than 1.2and up to 2.0 m	08-Aug-24	11-Aug-24	3500.00	4
 Depth up to 1.2 m			06-Aug-24	07-Aug-24		13
 11	AB-11	Depth up to 1.2 m	06-Aug-24	07-Aug-24	3000.00	13
 Supply, install, construct, test of pipe DN100 PN10			04-Aug-24	05-Aug-24		850
 10	AB-10	Supply, install, construct, test of pipe DN100 PN10	04-Aug-24	05-Aug-24	71.00	850
 Supply, install, construct, test of pipe DN200 PN10			31-Jul-24	01-Aug-24		2583
 9	AB-9	Supply, install, construct, test of pipe DN200 PN10	31-Jul-24	01-Aug-24	66.00	2583
 Supply, install, construct, test of air valves DN 80			29-Jul-24	30-Jul-24		1
 8	AB-8	Supply, install, construct, test of air valves DN 80	29-Jul-24	30-Jul-24	4000.00	1
 Supply, install, construct, test of check Valv DN100			25-Jul-24	28-Jul-24		2
 7	AB-7	Supply, install, construct, test of check Valv DN100	25-Jul-24	28-Jul-24	980.00	2
 Supply, install, construct, test of Gate Valve DN 100			23-Jul-24	24-Jul-24		4
 6	AB-6	Supply, install, construct, test of Gate Valve DN 100	23-Jul-24	24-Jul-24	650.00	4
 Supply, install, construct, test of pump 12 m3/h @Head 20 m(1 duty - 1 - standby)			21-Jul-24	22-Jul-24		2
 5	AB-5	Supply, install, construct, test of pump 12 m3/h @Head 20 m(1 duty - 1 - standby)	21-Jul-24	22-Jul-24	54750.00	2

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Page 6 of 6

6.4 Scheduling / Levelling Report

- The Scheduling/Levelling Report: in Primavera P6 is a detailed report that provides insights into the results of scheduling and resource levelling operations in a project schedule. It offers an overview of how scheduling adjustments or levelling actions impact the project timeline, resource allocation, and critical path. Here's a breakdown of what it includes and its purpose:
- Scheduling Information:
 - Start and Finish Dates: Shows the adjusted start and finish dates for activities based on the latest scheduling run.
 - Critical Path Analysis: Identifies which activities are on the critical path and highlights any changes, showing potential delays or shifts in critical activities.
 - Total Float/Slack: Shows the float values, indicating how much flexibility exists before an activity delay would affect the project end date. Activities with zero float are generally on the critical path.
 - Changes in Dependencies: Lists any modifications in activity relationships that may have resulted from the schedule run.
- Levelling Information:
 - Resource Allocation: Displays how resources are allocated after levelling, providing details on any changes made to reduce resource over-allocation (e.g., adjusting start dates for resource-heavy activities).
 - Levelling Delays: Reports any additional delays introduced due to resource levelling, as P6 adjusts start and finish dates to prevent resource conflicts.
 - Overallocated Resources: Identifies resources that remain overallocated after levelling, highlighting potential issues that need further adjustments.

- Purpose of the Report:
 - Progress Tracking: Helps you track changes to the project's schedule after running a schedule update or levelling. It's essential for determining if the project is moving forward as planned.
 - Optimization: Assists project managers in optimizing the schedule by balancing resource use across activities without extending the project end date unnecessarily.
 - Decision-Making: Supports decision-making by clearly showing where resources are stretched thin or where certain tasks might need adjustments, allowing for proactive schedule management.
- The Scheduling/Levelling Report is a valuable tool in project status meetings to communicate the impact of any schedule changes or levelling efforts, keeping the project aligned with its goals.
- Given in further pages are the reports generated through primavera for 4 phases.
- This report shows the progress done till each report was made.

REPORT 1

Scheduling/Levelling Report - 2024-07-04 14:05:32 - PM.EXE

=====

Default Project.....MTP

Use scheduling options from.....MTP

Projects:

MTP.....Construction of steel tank 3000

CUM capacity and Replacement of Existing Fiberglass tank in Mina

Scheduling/Levelling Settings:

General

SchedulingYes

LevellingNo

Ignore relationships to and from other projectsNo

Make open-ended activities criticalNo

Use Expected Finish DatesYes

Schedule automatically when a change affects datesNo

Level resources during schedulingNo

Recalculate assignment costs after schedulingNo

When scheduling progressed activities useRetained Logic

Calculate start-to-start lag fromEarly Start

Define critical activities as Total Float less than or equal to .0

Compute Total Float AsFinish Float

Calculate float based on finish date ofEach project

Calendar for scheduling Relationship LagPredecessor Activity

Calendar

Advanced

Calculate multiple float paths.....No

Statistics:

Projects.....1
Activities.....31
Not Started.....31
In Progress.....0
Completed.....0
Relationships.....30
Activities with Constraint.....0

Errors:

Warnings:

Activities without predecessors.....1
Project: MTP Activity: A Start

Activities without successors.....1
Project: MTP Activity: B finish

Out-of-sequence activities.....0

Activities with Actual Dates > Data Date.....0

Milestone Activities with invalid relationships.....0

Finish milestone and predecessors have different calendars.....0

Scheduling/Levelling Results:

Projects Scheduled/Levelled.....1
Activities Scheduled/Levelled.....31

Relationships with other projects.....0
 Data Date.....03-Jul-24
 Earliest Early Start Date.....03-Jul-24
 Latest Early Finish Date.....28-Oct-25

Exceptions:

Critical Activities.....31

Project:	MTP	Activity:	1	Taking the site from the client
Project:	MTP	Activity:	10	Process design calculations
Project:	MTP	Activity:	11	Approval of design
Project:	MTP	Activity:	12	Purchase order
Project:	MTP	Activity:	13	Manufacturing
Project:	MTP	Activity:	14	Delivery
Project:	MTP	Activity:	15	Foundation
Project:	MTP	Activity:	16	Placement of layer 1
Project:	MTP	Activity:	17	Fixing of layer 1
Project:	MTP	Activity:	18	Placement of layer 2
Project:	MTP	Activity:	19	Fixing of layer 2
Project:	MTP	Activity:	2	Surveying of site
Project:	MTP	Activity:	20	Placement of layer 3
Project:	MTP	Activity:	21	Fixing of layer 3
Project:	MTP	Activity:	22	Placement of layer 4
Project:	MTP	Activity:	23	Fixing of layer 4
Project:	MTP	Activity:	24	Placement of roof
Project:	MTP	Activity:	25	Fixing of roof
Project:	MTP	Activity:	26	Lighting system
Project:	MTP	Activity:	27	Construction of pipes and fittings
Project:	MTP	Activity:	28	Construction of fence work
Project:	MTP	Activity:	29	Primary hands over to the client
Project:	MTP	Activity:	3	Bore hole test for feasibility of
construction				
Project:	MTP	Activity:	4	Construction of rooms and
offices				

Project:	MTP	Activity:	5	Order for essentials for rooms and offices
Project:	MTP	Activity:	6	Delivery and setup of essentials for rooms and offices
Project:	MTP	Activity:	7	Layout drawing of the site
Project:	MTP	Activity:	8	Hydraulic loss calculations
Project:	MTP	Activity:	9	Hydraulic profile drawing
Project:	MTP	Activity:	A	Start
Project:	MTP	Activity:	B	finish

Activities with unsatisfied constraints.....0

Activities with unsatisfied relationships.....0

Activities with external dates.....0

REPORT 2

Scheduling/Levelling Report - 2024-07-04 14:37:08 - PM.EXE

=====

Default Project.....MTP

Use scheduling options from.....MTP

Projects:

MTP.....Construction of steel tank 3000

CUM capacity and Replacement of Existing Fiberglass tank in Mina

Scheduling/Levelling Settings:

General

SchedulingYes

LevellingNo

Ignore relationships to and from other projectsNo

Make open-ended activities criticalNo

Use Expected Finish DatesYes

Schedule automatically when a change affects datesNo

Level resources during schedulingNo

Recalculate assignment costs after schedulingNo

When scheduling progressed activities useRetained Logic

Calculate start-to-start lag fromEarly Start

Define critical activities as Total Float less than or equal to .0

Compute Total Float AsFinish Float

Calculate float based on finish date ofEach project

Calendar for scheduling Relationship LagPredecessor Activity

Calendar

Advanced

Calculate multiple float paths.....No

Statistics:

Projects.....1
Activities.....31
Not Started.....26
In Progress.....1
Completed.....4
Relationships.....30
Activities with Constraint.....0

Errors:

Warnings:

Activities without predecessors.....1
Project: MTP Activity: A Start

Activities without successors.....1
Project: MTP Activity: B finish

Out-of-sequence activities.....0

Activities with Actual Dates > Data Date.....0

Milestone Activities with invalid relationships.....0

Finish milestone and predecessors have different calendars.....0

Scheduling/Levelling Results:

Projects Scheduled/Levelled.....1
Activities Scheduled/Levelled.....31

Relationships with other projects.....0
 Data Date.....21-Aug-24
 Earliest Early Start Date.....21-Aug-24
 Latest Early Finish Date.....09-Nov-25

Exceptions:

Critical Activities.....27

Project:	MTP	Activity:	10	Process design calculations
Project:	MTP	Activity:	11	Approval of design
Project:	MTP	Activity:	12	Purchase order
Project:	MTP	Activity:	13	Manufacturing
Project:	MTP	Activity:	14	Delivery
Project:	MTP	Activity:	15	Foundation
Project:	MTP	Activity:	16	Placement of layer 1
Project:	MTP	Activity:	17	Fixing of layer 1
Project:	MTP	Activity:	18	Placement of layer 2
Project:	MTP	Activity:	19	Fixing of layer 2
Project:	MTP	Activity:	20	Placement of layer 3
Project:	MTP	Activity:	21	Fixing of layer 3
Project:	MTP	Activity:	22	Placement of layer 4
Project:	MTP	Activity:	23	Fixing of layer 4
Project:	MTP	Activity:	24	Placement of roof
Project:	MTP	Activity:	25	Fixing of roof
Project:	MTP	Activity:	26	Lighting system
Project:	MTP	Activity:	27	Construction of pipes and fittings
Project:	MTP	Activity:	28	Construction of fence work
Project:	MTP	Activity:	29	Primary hands over to the client
Project:	MTP	Activity:	4	Construction of rooms and
offices				
Project:	MTP	Activity:	5	Order for essentials for rooms
and offices				
Project:	MTP	Activity:	6	Delivery and setup of essentials
for rooms and offices				
Project:	MTP	Activity:	7	Layout drawing of the site

Project:	MTP	Activity:	8	Hydraulic loss calculations
Project:	MTP	Activity:	9	Hydraulic profile drawing
Project:	MTP	Activity:	B	finish

Activities with unsatisfied constraints.....0

Activities with unsatisfied relationships.....0

Activities with external dates.....0

REPORT 3

Scheduling/Levelling Report - 2024-07-04 14:43:01 - PM.EXE

=====

Default Project.....MTP

Use scheduling options from.....MTP

Projects:

MTP.....Construction of steel tank 3000

CUM capacity and Replacement of Existing Fiberglass tank in Mina

Scheduling/Levelling Settings:

General

SchedulingYes

LevellingNo

Ignore relationships to and from other projectsNo

Make open-ended activities criticalNo

Use Expected Finish DatesYes

Schedule automatically when a change affects datesNo

Level resources during schedulingNo

Recalculate assignment costs after schedulingNo

When scheduling progressed activities useRetained Logic

Calculate start-to-start lag fromEarly Start

Define critical activities as Total Float less than or equal to .0

Compute Total Float AsFinish Float

Calculate float based on finish date ofEach project

Calendar for scheduling Relationship LagPredecessor Activity

Calendar

Advanced

Calculate multiple float paths.....No

Statistics:

Projects.....1
Activities.....31
Not Started.....16
In Progress.....0
Completed.....15
Relationships.....30
Activities with Constraint.....0

Errors:

Warnings:

Activities without predecessors.....1
Project: MTP Activity: A Start

Activities without successors.....1
Project: MTP Activity: B finish

Out-of-sequence activities.....0

Activities with Actual Dates > Data Date.....0

Milestone Activities with invalid relationships.....0

Finish milestone and predecessors have different calendars.....0

Scheduling/Levelling Results:

Projects Scheduled/Levelled.....1
Activities Scheduled/Levelled.....31

Relationships with other projects.....0
 Data Date.....06-May-25
 Earliest Early Start Date.....06-May-25
 Latest Early Finish Date.....09-Nov-25

Exceptions:

Critical Activities.....16

Project:	MTP	Activity:	15	Foundation
Project:	MTP	Activity:	16	Placement of layer 1
Project:	MTP	Activity:	17	Fixing of layer 1
Project:	MTP	Activity:	18	Placement of layer 2
Project:	MTP	Activity:	19	Fixing of layer 2
Project:	MTP	Activity:	20	Placement of layer 3
Project:	MTP	Activity:	21	Fixing of layer 3
Project:	MTP	Activity:	22	Placement of layer 4
Project:	MTP	Activity:	23	Fixing of layer 4
Project:	MTP	Activity:	24	Placement of roof
Project:	MTP	Activity:	25	Fixing of roof
Project:	MTP	Activity:	26	Lighting system
Project:	MTP	Activity:	27	Construction of pipes and fittings
Project:	MTP	Activity:	28	Construction of fence work
Project:	MTP	Activity:	29	Primary hands over to the client
Project:	MTP	Activity:	B	finish

Activities with unsatisfied constraints.....0

Activities with unsatisfied relationships.....0

Activities with external dates.....0

REPORT 4

Scheduling/Levelling Report - 2024-07-07 08:48:17 - PM.EXE

=====

Default Project.....MTP

Use scheduling options from.....MTP

Projects:

MTP.....Construction of steel tank 3000

CUM capacity and Replacement of Existing Fiberglass tank in Mina

Scheduling/Levelling Settings:

General

SchedulingYes

LevellingNo

Ignore relationships to and from other projectsNo

Make open-ended activities criticalNo

Use Expected Finish DatesYes

Schedule automatically when a change affects datesNo

Level resources during schedulingNo

Recalculate assignment costs after schedulingNo

When scheduling progressed activities useRetained Logic

Calculate start-to-start lag fromEarly Start

Define critical activities as Total Float less than or equal to .0

Compute Total Float AsFinish Float

Calculate float based on finish date ofEach project

Calendar for scheduling Relationship LagPredecessor Activity

Calendar

Advanced

Calculate multiple float paths.....No

Statistics:

Projects.....1
Activities.....31
Not Started.....0
In Progress.....0
Completed.....31
Relationships.....30
Activities with Constraint.....0

Errors:

Warnings:

Activities without predecessors.....1
Project: MTP Activity: A Start

Activities without successors.....1
Project: MTP Activity: B finish

Out-of-sequence activities.....0

Activities with Actual Dates > Data Date.....0

Milestone Activities with invalid relationships.....0

Finish milestone and predecessors have different calendars.....0

Scheduling/Levelling Results:

Projects Scheduled/Levelled.....1
Activities Scheduled/Levelled.....31

Relationships with other projects.....0
Data Date.....10-Nov-25
Earliest Early Start Date.....10-Nov-25
Latest Early Finish Date.....10-Nov-25

Exceptions:

Critical Activities.....0

Activities with unsatisfied constraints.....0

Activities with unsatisfied relationships.....0

Activities with external dates.....0

7 Appendix

7.1 Internship confirmation

WASHNAH Contracting Co. LLC
SPECIALIZED IN WATER & WASTEWATER TREATMENT SYSTEMS
C.R.4030087384 - C.C.No.41998



شركة وشنه للمقاولات ذ.م.م
متخصصون في معالجة المياه والصرف الصحي والصناعي
س.ب.ت ٤٠٣٠٠٨٧٣٨٤ رقم العضوية : ٤١٩٩٨

Date: June 10, 2024
Ref #: WC-240610

Subject: Confirmation of Internship Request

Dear Mohammed Wafeeq Kazi,

We are pleased to inform you that your internship application at Washnah Contracting has been accepted. The internship will last for eight weeks and you will be assigned to work at our Projects & Engineering department, beginning on June 13, 2024.

During your internship, you will be required to report to our Jeddah office. Your working hours will be from 8:00 AM to 5:30 PM, Sunday through Thursday.

We are confident that this internship will provide you with valuable learning experiences and professional growth opportunities. We encourage you to make the most of this period by actively engaging with your assigned tasks and seeking guidance from your supervisors.

Should you have any questions or require further information before your start date, please feel free to contact our HR department.

We look forward to welcoming you to our team and wish you a successful and enriching internship.

Best regards,


Omar Almosallam
Human Resources Manager





ص.ب.ت ٥٠٧٤٠ جدة ٢١٥٣٣ - المملكة العربية السعودية - ت ٠١٢ ٦١٩٥٥١١ - بريد إلكتروني : info@washnah.com
P.O.Box: 50740, Jeddah 21533 - kingdom of Saudi Arabia - Tel : - 012 6195511 - www.washnah.com - Email : info@washnah.com

7.2 Internship certificate

WASHNAH Contracting Co. LLC SPECIALIZED IN WATER & WASTEWATER TREATMENT SYSTEMS C.R.4030087384 - C.C.No.41998		شركة وشنه للمقاولات ذ.م.م متخصصون في معالجة المياه والصرف الصحي والصناعي س.ت ٤٠٣٠٠٨٧٣٨٤ رقم العضوية : ٤١٩٩٨
--	---	--

Ref.# WC_240813	Date: Aug 13, 2024
-----------------	--------------------

INTERNSHIP CERTIFICATE

This is to certify that **Mr. Mohammed Wafeeq Mohd Salim Kazi**, student of Veermata Jijabai Technological Institute (VJI) has attended 8 weeks internship from 13-June-2024 up to 7-Aug-2024 in our project engineering and planning department

His details are given below:

Name	: MOHAMMED WAFEEQ MOHD SALIM KAZI
Date of Birth	: 22-APRIL-2003
Nationality	: INDIAN
Saudi Residency No.	: 2197354950
Indian Passport No	: S4653975

During the internship period he demonstrated a strong sense of responsibility, dedication and ability to understand and execute the task entitled to him.

We wish him good luck for his future studies and endeavors

For Washnah Contracting LLC

Omar Al Mosallam Human Resource Manager	 
--	--



ص.ب: ٥٠٧٤٠ جدة ٢١٥٣٣ - المملكة العربية السعودية - ت ٠١٢ ٦١٩٥٥١١ - بريد إلكتروني : info@washnah.com
P.O.Box: 50740 Jeddah 21533 - Kingdom of Saudi Arabia - Tel: 012 6295511 - www.washnah.com - Email: info@washnah.com

7.3 References

- Discharge Characteristics of Triangular-notch Thin-plate Weirs, By JOHN SHEN.
- CHEMICAL ENGINEERING HANDBOOK by PERRY & CHILTON.
- LAYMANS GUIDE BOOK FOR DEVELOPMENT OF SMALL HYDRO SITE by COMMISSION OF EUROPEAN COMUNITIES.
- WATER SUPPLY AND SEWERAGE 6th Edition by Terence J. McGhee.
- CAMEROON HYDRAULIC DATA by C.C. HEALD.